

# Multi-Cohort Transcriptomic Meta-Analysis of Glioblastoma Reveals a Convergent Immune-Myeloid Hub Network Associated with Adverse Prognosis and Druggable Targets

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## Abstract

**Background:** Glioblastoma (GBM) is the most lethal primary brain tumor in adults, yet transcriptomic heterogeneity across patient cohorts has severely limited the identification of reproducible prognostic biomarkers from single-cohort studies.

**Methods:** We performed an integrative multi-cohort meta-analysis of seven independent GBM transcriptomic datasets encompassing 680 samples across diverse clinical contexts, including survival stratification, treatment response, tumor recurrence, and tumor-versus-normal comparisons. Analytical methods included random-effects meta-analysis, protein–protein interaction (PPI) network construction, gene set enrichment analysis (GSEA), univariate Cox regression, LASSO-Cox and random forest machine learning, immune deconvolution via single-sample GSEA (ssGSEA), transcription factor enrichment, and drug target prioritization.

**Results:** Gene-level meta-analysis of 15,003 genes identified no genes reaching genome-wide significance ( $\text{FDR} < 0.05$ ), with pairwise  $\log_2$  fold-change correlations among survival cohorts uniformly weak (all  $r < 0.10$ ), confirming fundamental inter-cohort heterogeneity. However, PPI network analysis of the top 500 meta-analytic genes identified 52 hub genes centered on an immune-myeloid signaling cluster: SRC (degree = 91), H4C11 (62), CSF1R (56), APOE (54), CYBB (54), CD86 (52), BTK (51), and PIK3CG (50). GSEA revealed ribosomal pathway downregulation ( $\text{NES} = -2.33$ ,  $\text{padj} = 1.8 \times 10^{-8}$ ) and broad immune pathway suppression in short-survival GBM, with one carbon pool by folate as the sole significantly upregulated KEGG pathway ( $\text{NES} = 2.15$ ,  $\text{padj} = 5.5 \times 10^{-4}$ ). A 31-gene LASSO-Cox risk signature (C-index = 0.690; 95% CI 0.654–0.726) identified HOXC4, VAT1L, CAMKK2, and HMGA2 as top risk-associated features, converging with random forest classification. Immune deconvolution demonstrated that high-risk tumors are enriched for fibroblasts ( $r = 0.25$ ,  $\text{padj} = 2.5 \times 10^{-4}$ ), macrophages ( $r = 0.17$ ,  $\text{padj} = 0.019$ ), and neutrophils ( $r = 0.15$ ,  $\text{padj} = 0.044$ ). SPI1 (PU.1) was identified as the top master transcription factor regulator ( $\text{NES} = -2.22$ ,  $\text{padj} = 2.9 \times 10^{-5}$ ). Drug target analysis identified PIK3CG (130 drugs, 45 inhibitors), CSF1R (54 drugs, 22 inhibitors), BTK (52 drugs, 26 inhibitors), and SRC (54 drugs) as actionable hub targets.

**Conclusions:** While gene-level transcriptomic signals are poorly reproducible across GBM cohorts, network integration reveals a convergent immune-myeloid hub architecture that associates with prognosis through a fibroblast/macrophage-enriched tumor microenvironment, nominating CSF1R, BTK, and PIK3CG as druggable targets for immunomodulatory combination strategies.

**Keywords:** glioblastoma, meta-analysis, transcriptomics, immune microenvironment, myeloid signaling, SRC, CSF1R, tumor-associated macrophages, drug targets, prognostic signature

## Background

Glioblastoma (GBM; WHO grade 4 astrocytoma) is the most common and aggressive primary malignant brain tumor in adults, accounting for approximately 48% of all malignant central nervous system tumors with an annual incidence of approximately 3.2 per 100,000 person-years in the United States<sup>1</sup>. Despite decades of clinical investigation, the standard-of-care Stupp protocol—maximal safe surgical resection followed by

concurrent temozolomide chemotherapy and focal radiotherapy, with maintenance temozolomide—yields a median overall survival (OS) of only 14–16 months from diagnosis and a five-year survival rate below 5%<sup>2</sup>. The universal development of therapeutic resistance and the restricted penetration of systemic agents across the blood–brain barrier continue to frustrate therapeutic advances<sup>3</sup>.

A defining biological feature of GBM is its profound molecular heterogeneity, manifesting at the inter-tumoral level as distinct transcriptional subtypes—classified by The Cancer Genome Atlas (TCGA) as proneural, classical, and mesenchymal<sup>4;5</sup>—and at the intra-tumoral level as co-existing cellular states that transition dynamically<sup>6;7</sup>. This heterogeneity represents a fundamental obstacle to biomarker development: associations identified in single-cohort studies frequently fail to replicate across independent datasets. The *O*<sup>6</sup>-methylguanine-DNA methyltransferase (*MGMT*) promoter methylation status remains the only clinically validated predictive biomarker for temozolomide benefit in GBM<sup>8</sup>, yet its predictive value in specific subgroups remains debated<sup>3</sup>. There is therefore a critical and unmet need for additional prognostic frameworks grounded in mechanistic biology and validated across independent patient cohorts.

Meta-analytic integration of multiple independent cohorts provides a principled approach to this problem, aggregating statistical power while filtering cohort-specific technical noise. However, prior transcriptomic meta-analyses in GBM have generally been limited to comparisons between GBM and lower-grade gliomas or normal brain<sup>9</sup>, have relied on microarray-era datasets, or have focused on individual pathway analyses rather than integrating network topology, immune composition, regulatory circuits, and pharmacological actionability within a unified framework. In highly heterogeneous cancers such as GBM, gene-level meta-analysis alone often fails to identify reproducible biomarkers, motivating network-level and pathway-level analytical strategies<sup>10</sup>.

The tumor immune microenvironment (TME) of GBM is dominated by tumor-associated macrophages (TAMs) and microglia, which together constitute up to 40% of the tumor mass and impose a pervasively immunosuppressive milieu that blunts cytotoxic T-cell responses and correlates inversely with patient survival<sup>11;12</sup>. Recent work has highlighted the CSF1–CSF1R signaling axis as a key driver of TAM recruitment and polarization, with CSF1R inhibitors showing preclinical efficacy in GBM models<sup>13</sup>, although a phase II trial of the CSF1R inhibitor PLX3397 (pexidartinib) in recurrent GBM showed limited single-agent efficacy<sup>14</sup>. The myeloid transcription factor SPI1 (PU.1) serves as a master regulator of myeloid lineage commitment and has been shown to promote TAM-mediated glioma progression through BTK-dependent signaling<sup>15;16</sup>. PI3K $\gamma$  (encoded by *PIK3CG*), selectively expressed in leukocytes, functions as a molecular switch controlling immune suppression in the TME<sup>17</sup>. The SRC family kinases are central signaling nodes in both tumor cells and the TME, regulating invasion, migration, and immune cell activation<sup>18</sup>. Whether these immune-myeloid programs converge at the network level and associate with prognosis across independent GBM cohorts has not been comprehensively assessed.

Here we report a systematic integrative transcriptomic meta-analysis of seven independent GBM datasets ( $n = 680$  total samples) spanning diverse clinical contexts—survival

stratification, immunotherapy response, tumor recurrence, and tumor-versus-normal comparison—to capture the breadth of GBM transcriptomic biology. We deliberately selected datasets representing heterogeneous platforms, normalization strategies, and clinical endpoints for three reasons: (i) to test whether convergent biological signals emerge despite maximal technical and biological heterogeneity; (ii) to construct a comprehensive gene-level reference that spans multiple facets of GBM biology rather than a single clinical question; and (iii) to evaluate the limits of gene-level reproducibility in GBM and the necessity of network-level integration. Employing random-effects meta-analysis, PPI network construction, GSEA, immune deconvolution, LASSO-Cox and random forest machine learning, transcription factor enrichment, and drug target prioritization, we demonstrate that gene-level meta-analysis yields zero genome-wide significant hits, but that network integration reveals a convergent immune-myeloid hub architecture centered on SRC, CSF1R, BTK, and CD86 that associates with adverse prognosis through a fibroblast/macrophage-enriched TME. A 31-gene LASSO-Cox risk signature with HOXC4, CAMKK2, and HMGA2 as top features stratifies patients by survival. These findings prioritize CSF1R, BTK, and PIK3CG as druggable targets warranting investigation in immunomodulatory combination strategies for GBM.

## Methods

### Data Sources and Preprocessing

Seven publicly available GBM transcriptomic datasets were retrieved from the NCBI Gene Expression Omnibus (GEO) and the TCGA Genomic Data Commons (GDC) Data Portal (Table 1). GEO datasets were downloaded using the GEOquery Bioconductor package<sup>19</sup>. The cohorts were selected to encompass five distinct clinical contexts and data types: pre-normalized expression (GSE154041,  $n = 71$ ; short versus long OS in recurrent GBM), raw integer RNA-seq counts (GSE249282,  $n = 58$ , short versus long OS after dendritic cell vaccination; GSE186332,  $n = 57$ , immunotherapy responder versus non-responder; GSE79671,  $n = 35$ , short versus long OS), transcripts per million (TPM) values (GSE222515,  $n = 79$ ; primary versus recurrent GBM), DESeq2-normalized expression (GSE147352,  $n = 100$ ; GBM versus normal brain), and raw STAR-aligned counts from TCGA-GBM ( $n = 280$  primary tumor samples). For TCGA-GBM, samples were retrieved via the TCGAbiolinks package<sup>20</sup>; 87 multi-aliquot duplicate samples were removed, retaining the aliquot with highest library size per unique 16-character sample barcode. All samples represent primary GBM tumors, with the exception of GSE147352, which includes normal brain reference tissue retained solely for expression context.

The rationale for including datasets with heterogeneous contrasts (survival, treatment response, recurrence, tumor-versus-normal) is that the meta-analytic framework weights each study by its precision, and the network-level integration is designed to identify biological signals that are robust across diverse experimental conditions. Restricting to survival-only cohorts would have limited sample size and reduced the breadth of transcriptomic coverage.

Gene identifiers were harmonized to Ensembl gene IDs across all datasets. Entrez IDs in GSE186332 were converted using the org.Hs.eg.db annotation database. Gene-level filtering retained protein-coding genes expressed in at least two cohorts.

## Differential Expression Analysis

Differential expression was performed according to data type. For pre-normalized and TPM datasets (GSE154041, GSE222515, GSE147352), the limma-voom pipeline was applied following  $\log_2$  transformation and quantile normalization<sup>21</sup>. For raw count datasets (GSE249282, GSE186332, GSE79671, TCGA-GBM), DESeq2 with standard negative binomial modeling was used<sup>22</sup>. Batch effects in multi-platform cohort pairings were corrected using ComBat on  $\log_2$ -scale data<sup>23</sup> and ComBat-seq on raw counts<sup>24</sup>. The primary contrast for survival cohorts was short versus long overall survival, defined by median split of OS in months within each cohort. Surrogate variable analysis (SVA) was applied to control for latent technical confounders in cohorts with sufficient sample size<sup>25</sup>.

## Random-Effects Meta-Analysis

Gene-level  $\log_2$  fold-changes and standard errors from each study were integrated using random-effects meta-analysis with the restricted maximum likelihood (REML) estimator, implemented via the metafor R package<sup>26</sup>. Heterogeneity was quantified using the  $I^2$  statistic and Cochran's Q test. Pooled effect sizes, meta-analytic z-scores, and p-values were Benjamini-Hochberg corrected across all 15,003 tested genes<sup>27</sup>. Genes significant in only a single cohort ( $I^2$  undefined) were flagged to distinguish robust cross-cohort signals from single-dataset findings.

## Protein-Protein Interaction Network Analysis

A PPI network was constructed using STRINGdb (version 11.5; confidence threshold  $\geq 700$ )<sup>28</sup>, seeded with the top 500 genes ranked by random-effects meta-analytic p-value. Network topology was quantified using degree, betweenness centrality, closeness centrality, and PageRank metrics, as implemented in the igraph R package<sup>29</sup>. Hub genes were defined as nodes in the top 10% of the degree distribution (degree  $\geq 31$ ). Community detection used the Louvain modularity optimization algorithm.

## Gene Set Enrichment Analysis

Ranked gene lists derived from differential expression statistics (signed  $-\log_{10}$  p-value) were used as input for GSEA via the clusterProfiler R package (fgsea algorithm; 1,000 permutations)<sup>30;31</sup>. Gene sets tested included Gene Ontology Biological Process (GO-BP) and KEGG pathway databases. Enrichment results were filtered at FDR  $< 0.05$ . Normalized enrichment scores (NES) are reported with positive values indicating enrichment in the short-survival group and negative values indicating enrichment in the long-survival group.

## Transcription Factor Enrichment Analysis

Transcription factor (TF) target gene sets were obtained from the TRRUST v2 database. TF enrichment was assessed using the GSEA framework applied to TF-target gene sets, identifying master regulators whose targets are coordinately dysregulated in short-versus long-survival GBM.

## Immune Deconvolution

Relative immune cell infiltration was estimated using the single-sample GSEA (ssGSEA) algorithm implemented in the IOBR R package<sup>32;33</sup>. Scores were computed for ten immune cell populations: B cells, CD4<sup>+</sup> T cells, CD8<sup>+</sup> T cells, NK cells, dendritic cells, mast cells, monocytes, macrophages, neutrophils, and fibroblasts. Spearman rank correlation between LASSO-derived risk scores and immune infiltration scores was calculated; p-values were corrected by the Benjamini–Hochberg method.

## Univariate Cox Regression and Survival Analysis

Univariate Cox proportional hazards regression was performed for each hub gene in TCGA-GBM (n = 280) using overall survival data obtained from the GDC Clinical Data Portal<sup>20</sup>. Adjusted p-values were computed using the Benjamini–Hochberg procedure. Kaplan–Meier survival curves were constructed using the survminer R package and log-rank tests were applied for group comparisons.

## LASSO-Cox Regression and Risk Score Construction

Penalized Cox regression with LASSO ( $L_1$ ) regularization was applied using the glmnet R package with 10-fold cross-validation<sup>34;35</sup>. The regularization parameter  $\lambda$  was selected at the minimum cross-validated partial likelihood deviance ( $\lambda_{\min}$ ). The training cohort comprised GBM samples from the ComBat-corrected multi-cohort expression matrix (GSE154041 + GSE79671; n = 91). Risk scores were computed as the linear combination of LASSO-selected gene expression values weighted by their respective coefficients. Patients were stratified by the median risk score. External validation was performed in TCGA-GBM (n = 280), which was held out entirely from model fitting. Prognostic performance was assessed by Harrell’s C-index and time-dependent receiver operating characteristic (ROC) analysis<sup>36</sup>.

## Random Forest Classification

Random forest classification was performed to complement the LASSO-Cox approach, using a binary short/long survival outcome. The model was trained with 500 trees (ntree = 500), with the number of variables randomly sampled at each split set to  $\sqrt{p}$  (mtry =  $\sqrt{p}$ , where  $p$  is the number of features). Five-fold cross-validation was used to estimate generalization performance. Variable importance was assessed by mean decrease in accuracy and mean decrease in Gini index. Convergent genes identified by both LASSO and random forest were prioritized for biological interpretation.

## Drug Target Prioritization

Hub genes and differentially expressed genes were queried against the Drug–Gene Interaction Database (DGIdb v4.0) to identify pharmacological interactions<sup>37</sup>. Genes were annotated by gene family, druggability status, and count of known inhibitors. Integration of network hub status, differential expression direction, and druggability informed target prioritization.

## Cross-Cohort Validation

Cross-cohort reproducibility was assessed by computing Spearman rank correlations of  $\log_2$  fold-changes between each individual cohort and the meta-analytic reference, as well as pairwise LFC correlations among survival cohorts.

## Clinical Association Analysis

Risk score associations with clinical covariates (age at diagnosis, gender, race, survival group) were assessed in TCGA-GBM ( $n = 285$ ) using Spearman correlation (continuous variables) and Wilcoxon rank-sum or Kruskal–Wallis tests (categorical variables).

## Statistical Analysis

All analyses were performed in R (v4.3.0) with Bioconductor (v3.17). Significance was set at  $p_{adj} < 0.05$  unless otherwise specified. Visualizations were generated using ggplot2, survminer, pheatmap, and ggraph. All code is available at <https://github.com/hossainlab/GBM-Transcriptomic-MetaAnalysis>.

## Results

### Dataset Characteristics and Cross-Cohort Heterogeneity

Seven GBM transcriptomic datasets comprising 680 samples across diverse experimental platforms were analyzed (Table 1). Cohort sizes ranged from 35 samples (GSE79671) to 280 samples (TCGA-GBM). The datasets were selected to span five distinct clinical contexts: four cohorts employed short-versus-long OS as the primary contrast (GSE154041, GSE249282, GSE79671, TCGA-GBM), one compared immunotherapy responders with non-responders (GSE186332), one examined primary versus recurrent GBM (GSE222515), and one provided a GBM-versus-normal brain reference (GSE147352). This deliberate heterogeneity tests whether convergent transcriptomic signals emerge across fundamentally different experimental paradigms.

Cross-cohort concordance analysis among survival cohorts revealed striking heterogeneity (Figure 1). Pairwise  $\log_2$  fold-change correlations were uniformly weak: GSE154041 versus GSE79671 ( $r = 0.10$ ), GSE154041 versus TCGA-GBM ( $r = 0.08$ ), GSE79671 versus TCGA-GBM ( $r = -0.01$ ), and GSE249282 versus TCGA-GBM ( $r = -0.10$ ). Concordance with the meta-analytic reference varied substantially: GSE79671 exhibited the highest concordance ( $r = 0.82$ ), followed by GSE154041 ( $r = 0.69$ ), while TCGA-GBM ( $r = 0.01$ ) and GSE249282 ( $r = 0.02$ ) showed minimal concordance. This

inter-cohort heterogeneity, reflecting genuine biological differences between patient populations, treatment histories, molecular subtype compositions, and platforms, motivates the network-level and pathway-level analytical strategies employed in subsequent sections.

**Figure 1.** Cross-cohort concordance analysis. (A) Pairwise Spearman correlation heatmap of  $\log_2$  fold-changes among four survival cohorts ( $r$  range  $-0.10$  to  $0.10$ ). (B) Spearman correlations between each cohort and meta-analytic reference. (C–F) Per-cohort scatter plots of cohort-level versus meta-analytic  $\log_2$  fold-changes.

## Gene-Level Meta-Analysis: High Heterogeneity, No Genome-Wide Significant Hits

Random-effects meta-analysis across 15,003 genes identified zero genes reaching genome-wide significance at  $FDR < 0.05$  (Figure 2). The top candidates by raw p-value were TSFM (pooled  $\log_2FC = -3.21$ ,  $padj = 0.058$ ), CILP2 ( $\log_2FC = -4.22$ ,  $padj = 0.14$ ), and CSKMT ( $\log_2FC = -0.73$ ,  $padj = 0.14$ ). Notably, CSKMT and RGL1 ( $\log_2FC = 0.37$ ,  $padj = 0.23$ ) were among the few multi-study genes in the top-ranked candidates, with low heterogeneity ( $I^2 = 0$ ). The majority of top-ranked meta-analytic findings were attributable to single-cohort signals ( $I^2$  undefined), with only 6 of the top 20 genes showing evidence from multiple studies.

The absence of genome-wide significant genes is consistent with the weak pairwise LFC correlations observed across survival cohorts and reflects the fundamental transcriptional heterogeneity of GBM. This result is itself informative: it demonstrates that individual gene-level biomarkers cannot be reliably identified across independent GBM cohorts using standard meta-analytic approaches, directly motivating the network-level integration that constitutes the primary analytical contribution of this study.

**Figure 2.** Meta-analysis volcano plot. Random-effects meta-analysis results for 15,003 genes displayed as pooled  $\log_2$  fold-change versus  $-\log_{10}(padj)$ . Dashed horizontal line indicates  $FDR = 0.05$  significance threshold. No genes cross this threshold. Top candidates labeled: TSFM, CILP2, CSKMT, RGL1.

## PPI Network Reveals Immune-Myeloid Hub Architecture

A PPI network constructed from the top 500 meta-analytic genes comprised 483 nodes connected by 3,534 high-confidence interactions (STRING confidence  $\geq 700$ ; Figure 3). The top 10% of nodes by degree centrality defined 52 hub genes (Table 4). The highest-degree hub gene was SRC (degree = 91, betweenness = 0.070), followed by H4C11 (degree = 62), CSF1R (degree = 56), APOE (degree = 54), CYBB (degree = 54), PPP1CA (degree = 54), CD86 (degree = 52), BTK (degree = 51), PIK3CG (degree = 50), and TYROBP (degree = 48).

**Figure 3.** PPI network of top 500 meta-analytic genes. Nodes represent genes, edges represent high-confidence STRING interactions (confidence  $\geq 700$ ). Node size proportional to degree centrality. Hub genes (top 10% by degree) highlighted.



The hub gene set was strikingly enriched for immune and myeloid signaling molecules (Figure 4). CSF1R, CD86, BTK, TYROBP, and CYBB constitute a coherent myeloid/macrophage signaling cluster, representing the functional myeloid infiltrate within the tumor bed. The signal transduction hubs SRC, PIK3CG, and PPP1CA serve as central integrators linking receptor signaling to downstream effector pathways in both tumor cells and immune cells (Figure 4). Louvain community detection identified seven network modules. Module 1 was dominated by ribosomal proteins (RPL7A, RPS9, RPS13, RPL9, RPL32, RPLP0, RPLP2, RPL26L1, RPL27) and translation-associated genes (PRPF8, QARS1, MRPL2, SF3B5), all exhibiting negative  $\log_2FC$  values indicating downregulation in short-survival GBM.

**Figure 4.** Hub gene degree centrality barplot. Top 52 hub genes ranked by degree centrality, colored by functional category. Immune-myeloid genes dominate the highest-degree positions.

## Pathway Enrichment: Ribosomal Downregulation and Immune Suppression

GSEA identified 91 significantly enriched KEGG pathways and 1,044 GO-BP terms at  $FDR < 0.05$ , revealing a coherent biological program differentiating short- from long-survival GBM (Figure 5; Table 3).

**Figure 5.** GSEA enrichment results. (A) KEGG pathway dotplot showing top enriched and depleted pathways. Ribosome (NES =  $-2.33$ ) is most depleted; One carbon pool by folate (NES =  $2.15$ ) is sole significantly enriched positive pathway. (B) GO-BP enrichment dotplot showing top biological process terms.

The most significantly depleted KEGG pathway in short-survival tumors was Ribosome (hsa03010; NES =  $-2.33$ ,  $padj = 1.8 \times 10^{-8}$ ), followed by Coronavirus disease (NES =  $-2.25$ ,  $padj = 1.8 \times 10^{-8}$ ), Staphylococcus aureus infection (NES =  $-2.24$ ,  $padj = 2.4 \times 10^{-5}$ ), Systemic lupus erythematosus (NES =  $-2.12$ ,  $padj = 2.5 \times 10^{-5}$ ), Hematopoietic cell lineage (NES =  $-2.09$ ,  $padj = 1.1 \times 10^{-4}$ ), and Cytokine-cytokine receptor interaction (NES =  $-2.07$ ,  $padj = 2.5 \times 10^{-5}$ ). Complement and coagulation cascades (NES =  $-2.01$ ,  $padj = 5.8 \times 10^{-4}$ ) and multiple immune-related pathways were also significantly depleted. The negative enrichment of infection-related KEGG pathways (Staphylococcus aureus, Coronavirus disease, Leishmaniasis) reflects the shared innate immune gene content of these gene sets rather than literal infectious etiology.

Only one KEGG pathway was significantly enriched in short-survival GBM: One carbon pool by folate (hsa00670; NES =  $2.15$ ,  $padj = 5.5 \times 10^{-4}$ ; Figure 6).

**Figure 6.** GSEA running enrichment score plots for top KEGG pathways: Ribosome, Staphylococcus aureus infection, One carbon pool by folate, and Cytokine-cytokine receptor interaction.

At the GO-BP level, the top depleted terms were dominated by immune and cell fusion processes: regulation of syncytium formation (NES =  $-2.37$ ), regulation of myoblast fusion (NES =  $-2.35$ ), leukocyte proliferation (NES =  $-2.24$ ), positive regulation of

leukocyte cell–cell adhesion (NES =  $-2.17$ ), and interleukin-6 production (NES =  $-2.16$ ). Collectively, these results indicate that short-survival GBM is characterized by a dual program of ribosomal/translation machinery downregulation and immune pathway suppression.

## Transcription Factor Analysis Identifies SPI1 as Master Regulator

Transcription factor enrichment analysis identified SPI1 (PU.1) as the most significantly enriched master regulator (NES =  $-2.22$ ,  $padj = 2.9 \times 10^{-5}$ ; Figure 7).

**Figure 7.** Transcription factor enrichment analysis. SPI1/PU.1 is most significantly enriched TF (NES =  $-2.22$ ,  $padj = 2.9 \times 10^{-5}$ ), followed by ETS1, RFX5, and EPAS1.

SPI1 targets were coordinately downregulated in short-survival GBM. ETS1 (NES =  $-2.02$ ,  $padj = 8.8 \times 10^{-4}$ ), RFX5 (NES =  $-1.88$ ,  $padj = 0.035$ ), and EPAS1/HIF-2 $\alpha$  (NES =  $-1.83$ ,  $padj = 0.016$ ) were also significantly enriched.

The convergent identification of SPI1 as the top regulatory TF and myeloid genes as dominant network hubs provides orthogonal evidence for a myeloid transcriptional axis in GBM prognosis.

## Survival Analysis and Machine Learning Risk Model

### Univariate Cox Regression

Univariate Cox proportional hazards regression of hub genes in TCGA-GBM ( $n = 280$ ) identified no genes reaching statistical significance at  $padj < 0.05$  (Table 2). The top nominally significant genes (unadjusted  $p < 0.05$ ) included HMGA2 (HR = 1.09, 95% CI 1.04–1.15,  $p = 0.001$ ,  $padj = 0.11$ ), CCR7 (HR = 1.18, 95% CI 1.06–1.32,  $p = 0.002$ ,  $padj = 0.11$ ), HOXC4 (HR = 1.13, 95% CI 1.03–1.25,  $p = 0.009$ ,  $padj = 0.23$ ), CAMKK2 (HR = 1.33, 95% CI 1.06–1.66,  $p = 0.012$ ,  $padj = 0.23$ ), SDC1 (HR = 1.11, 95% CI 1.02–1.20,  $p = 0.013$ ,  $padj = 0.23$ ), DLG4 (HR = 1.25, 95% CI 1.04–1.51,  $p = 0.017$ ,  $padj = 0.23$ ), and SERPINE1 (HR = 1.10, 95% CI 1.02–1.19,  $p = 0.018$ ,  $padj = 0.23$ ). Two ribosomal genes showed nominally protective associations: RPL7A (HR = 0.81,  $p = 0.020$ ) and RPL32 (HR = 0.81,  $p = 0.020$ ).

The absence of FDR-significant prognostic genes from univariate analysis is consistent with the gene-level heterogeneity observed in the meta-analysis and underscores the value of multivariate machine learning approaches for prognostic modeling in a disease with profound inter-patient heterogeneity.

### LASSO-Cox Risk Signature

LASSO-Cox regression applied to the multi-cohort training matrix ( $n = 91$ ) selected 31 genes at  $\lambda_{\min}$  (Table 5; Figure 8).

**Figure 8.** LASSO-Cox prognostic signature. (A) Cross-validation curve showing mean deviance versus  $\log(\lambda)$ ; 31 genes selected at  $\lambda_{\min}$ . (B) Coefficient plot for the 31-gene signature.

Among risk-associated genes (positive coefficient), the top features were HOXC4 (coefficient = 0.25), VAT1L (0.18), CAMKK2 (0.17), HMGA2 (0.16), SDC1 (0.11), TSFM (0.11), RIC8A (0.08), PNPLA7 (0.08), BHMT2 (0.08), PCDHA11 (0.06), and TNFRSF12A (0.05). Among protective genes (negative coefficient), the top features were QARS1 (−0.20), ZKSCAN8 (−0.16), CD207 (−0.13), POU4F1 (−0.09), NPY1R (−0.09), CD86 (−0.05), CAV1 (−0.05), and CSKMT (−0.05).

Risk scores computed from the 31-gene signature demonstrated the expected gradient across survival groups in TCGA-GBM: long-survival patients had a mean risk score of −0.30 (SD = 0.42), medium-survival patients 0.03 (SD = 0.42), and short-survival patients 0.13 (SD = 0.52; Figure 9). Prognostic discrimination was assessed in the held-out TCGA-GBM validation cohort (n = 283): Harrell’s C-index was 0.690 (95% CI 0.654–0.726,  $p = 1.3 \times 10^{-24}$ ), and time-dependent AUC values were 0.73 (95% CI 0.66–0.79) at 12 months, 0.80 (95% CI 0.74–0.85) at 18 months, and 0.79 (95% CI 0.73–0.85) at 24 months, indicating good discriminative ability.

**Figure 9.** Risk score validation. **(A)** Distribution of 31-gene risk scores across TCGA-GBM patients ordered by risk, with survival and vital status. **(B)** Kaplan–Meier curves for high- versus low-risk groups.

### Random Forest Classification

Random forest classification (ntree = 500, mtry =  $\sqrt{p}$ , 70/30 train-test split) achieved an out-of-bag (OOB) classification error of 37.6% and a test set AUC of 0.55 (95% CI 0.39–0.71), with five-fold cross-validated AUC of 0.55 (95% CI 0.47–0.63) for discriminating short- from long-survival patients. The modest RF discriminative performance—in contrast to the stronger LASSO-Cox results (C-index = 0.690)—likely reflects the fundamental difficulty of binary classification in a disease where survival is continuous and subtype boundaries are blurred. The top features by mean decrease in Gini index were TOGARAM2, HOXC4, MYO1F, COL8A2, and PLCH2. Notably, HOXC4 appeared among the top features in both the LASSO-Cox and random forest models, along with BHMT2 and HMGA2. These convergent genes—identified independently by two complementary machine learning algorithms with fundamentally different optimization objectives—represent the most robust prognostic candidates and are prioritized for biological interpretation.

### Immune Microenvironment Composition

Immune deconvolution by ssGSEA in TCGA-GBM (n = 285) revealed three immune cell populations with significant associations to the 31-gene risk score after multiple testing correction (Table 6; Figure 10).

**Figure 10.** Immune microenvironment landscape. **(A)** Heatmap of ssGSEA immune cell infiltration scores across 285 TCGA-GBM samples ordered by risk score. **(B)** Spearman correlation barplot between risk score and immune cell types. Fibroblast ( $r = 0.25$ ), macrophage ( $r = 0.17$ ), and neutrophil ( $r = 0.15$ ) reach significance.

The strongest association was with fibroblast content (Spearman  $r = 0.25$ ,  $p_{adj} = 2.5 \times 10^{-4}$ ), followed by macrophage content ( $r = 0.17$ ,  $p_{adj} = 0.019$ ) and neutrophil content ( $r = 0.15$ ,  $p_{adj} = 0.044$ ). CD4<sup>+</sup> T cells ( $r = 0.12$ ,  $p_{adj} = 0.12$ ), monocytes ( $r = 0.11$ ,  $p_{adj} = 0.12$ ), CD8<sup>+</sup> T cells ( $r = 0.09$ ,  $p_{adj} = 0.21$ ), and mast cells ( $r = -0.06$ ,  $p_{adj} = 0.31$ ) did not reach significance after correction.

The fibroblast/macrophage/neutrophil enrichment in high-risk tumors is consistent with a mesenchymal, myeloid-dominated TME phenotype. This finding converges with the network hub analysis (SRC, CSF1R, CD86, BTK, TYROBP as myeloid hubs) and the TF analysis (SPI1 as master myeloid regulator) to support a model in which aggressive GBM harbors a fibroblast-rich, macrophage-infiltrated microenvironment driven by myeloid transcriptional programs.

## Clinical Associations

The risk score correlated positively with patient age at diagnosis (Spearman  $r = 0.23$ ,  $p = 9.1 \times 10^{-5}$ ,  $n = 285$ ; Figure 11), consistent with the well-established adverse prognostic effect of older age in GBM. Gender ( $p = 0.38$ , Wilcoxon test) and race ( $p = 0.80$ , Kruskal–Wallis test) were not significantly associated with risk score.

**Figure 11.** Clinical association plots. (A) Correlation between risk score and age at diagnosis (Spearman  $r = 0.23$ ,  $p = 9.1 \times 10^{-5}$ ). (B) Risk scores by gender ( $p = 0.38$ ). (C) Risk scores by race ( $p = 0.80$ ). (D) Risk scores by survival group stratification.

## Drug Target Prioritization

Among 69 druggable genes in the analysis, 1,265 drug–gene interactions were identified from DGIdb (Figure 12; Table 7). Hub genes with the highest druggability included PIK3CG (130 drugs, 45 inhibitors; phosphatidylinositol 3-kinase family), SRC (54 drugs, 16 inhibitors; tyrosine kinase), CSF1R (54 drugs, 22 inhibitors; receptor tyrosine kinase), and BTK (52 drugs, 26 inhibitors; tyrosine kinase). All four are clinically actionable targets with approved or investigational inhibitors.

**Figure 12.** Drug target prioritization. (A) Lollipop plot of druggable hub genes with DGIdb interactions, displayed as number of drug interactions colored by differential expression direction. (B) Gene family annotation barplot.

No targets achieved “High” priority status (defined as hub + significantly differentially expressed + significant Cox HR). Seven hub genes achieved “Medium” priority: PIK3CG, GRIN2A, APOE, CSF1R, SRC, BTK, and SERPINE1. The absence of high-priority targets reflects the stringent requirement for convergent statistical significance across all three criteria and is consistent with the gene-level heterogeneity observed throughout this study; the medium-priority targets nonetheless represent biologically motivated candidates supported by network topology, directional expression change, and pharmacological accessibility.

## Discussion

This study presents a systematic multi-cohort integrative analysis of GBM transcriptomics across seven independent datasets totaling 680 samples, representing, to our knowledge, one of the most comprehensive network-based transcriptomic meta-analyses of GBM to date. The central finding is that while gene-level meta-analysis fails to identify reproducible biomarkers across GBM cohorts (0 genes at  $FDR < 0.05$ ), network-level integration reveals a convergent immune-myeloid signaling hub centered on SRC, CSF1R, BTK, and CD86 that associates with prognosis through a fibroblast/macrophage-enriched TME.

## Study Rationale and Contribution

The rationale for this study rests on three observations. First, single-cohort GBM transcriptomic studies have proliferated but suffer from limited sample sizes, lack of cross-cohort validation, and susceptibility to false-positive findings driven by cohort-specific confounders. Second, prior meta-analyses of GBM have largely relied on microarray data, focused on GBM-versus-normal or GBM-versus-lower-grade comparisons<sup>9</sup>, or have not integrated network topology with immune profiling and drug target analysis in a unified computational framework. Third, the failure of gene-level meta-analysis to produce significant hits in GBM—a finding we confirm and quantify—mandates network-level analytical strategies that can detect convergent biological programs operating through multiple genes.

Our contribution extends beyond prior work in several respects. We demonstrate that the inter-cohort heterogeneity in GBM is so profound that pairwise LFC correlations among survival cohorts are uniformly weak ( $r < 0.10$ ), a quantitative characterization that has not been previously reported across this number of RNA-seq cohorts. We show that despite this gene-level discordance, a convergent immune-myeloid hub network emerges from PPI analysis of top meta-analytic genes, and that this network is independently corroborated by immune deconvolution, transcription factor analysis, and drug target mapping. The integration of LASSO-Cox and random forest machine learning with convergent gene identification (HOXC4, CAMKK2, HMGA2) provides a multi-algorithm validated prognostic signature. Finally, the drug target prioritization connects biological findings directly to actionable pharmacology, bridging discovery and translational potential.

## Gene-Level Heterogeneity as Fundamental GBM Biology

The absence of genome-wide significant genes from meta-analysis of 15,003 genes across four survival cohorts is not a negative result but rather reflects the fundamental transcriptomic heterogeneity of GBM. Pairwise LFC correlations between survival cohorts were uniformly weak ( $r$  range  $-0.10$  to  $0.10$ ), indicating that even cohorts studying the same biological question (short versus long OS) produce largely discordant gene-level signatures. This discordance likely reflects multiple sources of biological and technical variation: distinct molecular subtype compositions across cohorts<sup>4</sup>, heterogeneous treatment histories, variable tumor cellularity, platform-specific biases,

and the stochastic nature of transcriptomic sampling from a profoundly heterogeneous disease with co-existing cellular states<sup>6;38;39</sup>. Our results validate the view that single-gene biomarker strategies are fundamentally limited in GBM and that pathway-level and network-level approaches are necessary to extract reproducible biological signals<sup>10;40</sup>. While some studies have proposed individual prognostic markers<sup>41;42</sup>, our meta-analysis suggests these findings may often be cohort-specific or susceptible to low statistical power in single-cohort settings<sup>43;44</sup>.

## SRC-Centered Myeloid Hub Network

SRC emerged as the dominant hub gene (degree = 91), substantially exceeding the next-highest hub (H4C11, degree = 62). SRC family kinases are multifunctional non-receptor tyrosine kinases that regulate cell proliferation, survival, migration, and invasion in multiple cancer types, including GBM<sup>18;45</sup>. In the GBM context, SRC has been implicated in EGFR-driven invasion, integrin-mediated adhesion, and STAT3-dependent immune evasion<sup>46–48</sup>. The positioning of SRC as a central network hub—connecting myeloid signaling nodes (CSF1R, BTK, TYROBP) with signal transduction components (PIK3CG, PPP1CA) and structural/adhesion molecules—suggests that SRC serves as a molecular integrator of tumor-TME crosstalk<sup>49;50</sup>.

The CSF1R/CD86/BTK/TYROBP/CYBB cluster constitutes a coherent TAM signature<sup>51;52</sup>. The CSF1–CSF1R axis is the principal driver of macrophage recruitment and M2 polarization in the tumor bed, and CSF1R-expressing TAMs promote tumor invasion and immune suppression<sup>12;13</sup>. CD86, as a co-stimulatory ligand on antigen-presenting cells<sup>53</sup>, and BTK, as a kinase essential for myeloid cell activation and Akt/mTOR pathway induction<sup>15</sup>, further define this cluster as representing the functional myeloid infiltrate. TYROBP (DAP12) and CYBB (gp91-phox) act as key adaptors and effector enzymes, respectively, in myeloid activating receptors and oxidative stress responses<sup>54;55</sup>. The coordinate appearance of these genes as network hubs provides systems-level evidence for the central role of myeloid cells in the GBM interaction network<sup>56;57</sup>.

APOE (degree = 54) and H4C11 (62) represent additional hub signals linked to lipid-mediated immunosuppression and epigenetic remodeling, respectively<sup>58;59</sup>. APOE expression by TAMs has been shown to drive glioma progression by modulating CCL5/CCR5 signaling and inducing M2 polarization, consistent with our identification of an immune-myeloid axis.

## Immune-Myeloid Axis in GBM Prognosis

The convergent evidence from PPI network hubs (myeloid genes), immune deconvolution (fibroblast/macrophage/neutrophil enrichment), and master TF analysis (SPI1) supports a model in which aggressive GBM harbors a myeloid-dominated, immunosuppressive TME<sup>60;61</sup>. The fibroblast enrichment in high-risk tumors ( $r = 0.25$ ,  $p_{adj} = 2.5 \times 10^{-4}$ ) is consistent with the mesenchymal GBM subtype, characterized by extensive ECM remodeling<sup>4;62</sup>. Recent studies have highlighted CAFs as active components that recruit M2-like macrophages and promote therapy resistance<sup>42;63;64</sup>. The

macrophage association ( $r = 0.17$ ,  $padj = 0.019$ ) corroborates the extensive literature on TAM-mediated immunosuppression<sup>11;12</sup>.

SPI1 (PU.1), identified as the top master TF regulator ( $NES = -2.22$ ,  $padj = 2.9 \times 10^{-5}$ ), is the transcriptional master switch for myeloid lineage commitment<sup>65;66</sup>. The downregulation of SPI1 targets in short-survival GBM may appear paradoxical given the macrophage enrichment, but is consistent with a model in which TAMs in aggressive GBM adopt a dysfunctional, immunosuppressive phenotype with attenuated canonical activation programs<sup>16;67;68</sup>. Recent evidence directly demonstrates that PU.1 induces TAM polarization promoting glioma progression through BTK-mediated signaling<sup>15</sup>, connecting the TF and network findings.

The ribosomal pathway downregulation ( $NES = -2.33$ ) and depletion of leukocyte proliferation ( $NES = -2.24$ ) observed in GSEA suggest translational stress and systemic immune derangement as additional features of aggressive GBM<sup>69;70</sup>. Ribosomal biogenesis suppression has been linked to therapy resistance and cancer metastasis through p53-independent stress responses<sup>71;72</sup>. One-carbon metabolism ( $NES = 2.15$ ) appears as a metabolic vulnerability in short-survival GBM, supporting nucleotide biosynthesis in proliferating tumor cells<sup>73-75</sup>.

## Multi-Gene Risk Signature: Biological Interpretation

The 31-gene LASSO-Cox signature captures the multi-dimensional biology of a disease as heterogeneous as GBM more effectively than single-gene models. The methodological robustness of this signature is supported by previous studies employing multi-algorithm (Cox + LASSO + RF) feature selection in glioma<sup>76-78</sup>. The top risk-associated genes provide mechanistic insight into the biology of aggressive GBM.

HOXC4 (coefficient = 0.25) is a developmental transcription factor identified as a top risk gene in pediatric and adult gliomas and pan-cancer analyses<sup>79-81</sup>. CAMKK2 (coefficient = 0.17) promotes immunosuppression and checkpoint blockade resistance in GBM, maintaining myeloid cells in a disease-associated microglial-like phenotype<sup>82;83</sup>. HMGA2 (coefficient = 0.16) is a validated oncogenic driver that promotes stemness and tumorigenicity, particularly in CD133<sup>+</sup> neurosphere cells<sup>84;85</sup>. VAT1L (0.18) correlates with immunosuppressive actions and radioresistance in diffuse gliomas, consistent with our signature's phenotype<sup>86;87</sup>.

Among protective genes, CD207/Langerin (coefficient =  $-0.13$ ) and CD86 ( $-0.05$ ) may reflect specific dendritic cell or antigen-presenting cell populations associated with a more favorable TME. QARS1 (coefficient =  $-0.20$ ), a glutaminyl-tRNA synthetase, may reflect intact translational machinery as a marker of less aggressive biology, consistent with the ribosomal downregulation signature associated with poor prognosis.

The convergence of HMGA2, HOXC4, and CAMKK2 across univariate Cox regression, LASSO-Cox, and random forest importance represents the strongest prognostic signal from our analysis and prioritizes these genes for functional validation.

## Therapeutic Implications

The druggability analysis identifies several actionable targets at the intersection of hub gene status and pharmacological accessibility.

CSF1R inhibitors represent a TAM-depleting strategy with strong biological rationale. Pexidartinib (PLX3397), an FDA-approved CSF1R inhibitor, has been investigated in a phase II trial in recurrent GBM, where it demonstrated adequate CNS penetration but limited single-agent efficacy<sup>14</sup>. CSF1R inhibition has been shown to abrogate microenvironment-mediated therapeutic resistance in glioma models<sup>88</sup>, suggesting that combination approaches with immune checkpoint inhibitors or standard chemotherapy may be required to realize the therapeutic potential of TAM modulation<sup>13</sup>.

BTK inhibitors, including ibrutinib and acalabrutinib, are approved for lymphoid malignancies with documented CNS penetration. Preclinical investigation has demonstrated that ibrutinib suppresses glioma tumorigenesis and stem cell phenotypes<sup>89</sup>, and BTK is expressed in both B cells and myeloid cells, positioning BTK inhibition as a strategy to modulate TAM polarization in GBM.

PIK3CG (PI3K $\gamma$ ) is selectively expressed in leukocytes and functions as a molecular switch controlling immune suppression versus activation in the TME<sup>17</sup>. PI3K $\gamma$  inhibition has shown synergy with immune checkpoint blockade in preclinical models, making it a compelling candidate for combination immunotherapy in GBM.

SRC inhibitors, including dasatinib, have been evaluated in GBM clinical trials with limited single-agent efficacy but remain candidates for combination strategies targeting both tumor-intrinsic and TME-mediated resistance.

The convergence of multiple druggable hub genes within the myeloid signaling axis motivates a combination therapeutic strategy targeting the CSF1R/BTK/PIK3CG/SRC network to reprogram the immunosuppressive GBM microenvironment.

## Limitations

Several limitations should be acknowledged. First, all data derive from bulk RNA sequencing, which cannot resolve intra-tumoral heterogeneity that single-cell approaches can address. Second, heterogeneous platforms and normalization methods across cohorts contribute to the inter-cohort discordance observed in the meta-analysis; while this heterogeneity was deliberate (to test robustness), it also increases noise and reduces power for gene-level detection. Third, no genes reached FDR < 0.05 in either the meta-analysis or univariate Cox regression, limiting gene-level prognostic conclusions to nominal significance and network-level inferences. Fourth, the LASSO-Cox model was trained on a relatively small cohort (n = 91), which may limit its generalizability; prospective validation in larger, standardized cohorts is essential. Fifth, ssGSEA provides relative, not absolute, immune cell quantification and cannot substitute for histological or flow cytometric assessment. Sixth, the study is retrospective and all findings require prospective validation in independent cohorts with standardized expression profiling and clinical annotation. Seventh, the inclusion of non-survival contrasts (recurrence, response, tumor-versus-normal) in the meta-analysis introduces



heterogeneity in the biological question being asked, though this is mitigated by the random-effects framework that accommodates between-study variation. Eighth, the analytical pipeline is sequential rather than fully independent: the top 500 meta-analytic genes seed the PPI network, from which hub genes are derived and subsequently used in GSEA and survival analysis. While the transcription factor enrichment and immune deconvolution analyses provide partially independent corroboration, the network-derived findings should be interpreted as hypothesis-generating rather than as independent validation. Ninth, *MGMT* promoter methylation status and *IDH* mutation status—the two most established molecular prognostic markers in GBM—were not available for all cohorts and were not incorporated as covariates in the meta-analysis or survival models. The prognostic associations reported here may therefore be confounded by or partially attributable to *MGMT* or *IDH* status variation across cohorts. Future studies should explicitly stratify by these molecular markers to isolate the independent prognostic contribution of the immune-myeloid signatures identified here. Finally, the one carbon pool by folate enrichment in short-survival GBM, while statistically significant, represents a single pathway finding that requires independent confirmation and mechanistic investigation.

## Conclusions

Multi-cohort GBM meta-analysis reveals that gene-level transcriptomic signals are poorly reproducible across independent cohorts, but network integration identifies a convergent immune-myeloid hub architecture centered on *SRC*, *CSF1R*, *BTK*, and *CD86*. High-risk tumors harbor fibroblast- and macrophage-enriched microenvironments consistent with a mesenchymal, myeloid-dominated phenotype, driven by *SPI1*-regulated transcriptional programs. A 31-gene LASSO-Cox risk signature with *HOXC4*, *CAMKK2*, and *HMGA2* as top features stratifies patients by survival, with convergent identification across two independent machine learning algorithms. Ribosomal pathway downregulation and immune pathway suppression characterize the transcriptomic landscape of aggressive GBM. These findings prioritize *CSF1R*, *BTK*, and *PIK3CG* as druggable targets warranting investigation in GBM, with combination strategies targeting the myeloid-immune axis representing a rational therapeutic approach grounded in network biology.

## Declarations

### Ethics approval and consent to participate

Not applicable. This study used only publicly available, de-identified datasets from the Gene Expression Omnibus and the TCGA Genomic Data Commons. No new human subjects research was conducted.

### Consent for publication

Not applicable.

## Availability of data and materials

All GEO datasets analyzed in this study are publicly available: GSE154041, GSE249282, GSE186332, GSE222515, GSE79671, and GSE147352. TCGA-GBM data were accessed via the NCI Genomic Data Commons at <https://portal.gdc.cancer.gov>. All analysis code and processed data matrices are available at <https://github.com/hossainlab/GBM-Transcriptomic-MetaAnalysis>.

## Competing interests

The authors declare that they have no competing interests.

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## Authors' contributions

MJH conceived and designed the study, developed the analytical pipeline, performed all computational analyses, created visualizations, and wrote the original draft. FTJF, EA, RHS, JI, MHC, MLA, SD, and MMH contributed to validation and reviewed and edited the manuscript. All authors read and approved the final manuscript.

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## Tables

**Table 1. Overview of GBM Transcriptomic Datasets Analyzed**

**Table 1:** OS = overall survival; CPM = counts per million; TPM = transcripts per million; GEO = Gene Expression Omnibus; GDC = Genomic Data Commons; DC = dendritic cell. **Table 2. Univariate Cox Proportional Hazards Regression Results for Top Hub Genes in TCGA-GBM**

| Dataset   | Source | Data Type            | n  | Primary Contrast  | DEA Method | Rationale                                |
|-----------|--------|----------------------|----|-------------------|------------|------------------------------------------|
| GSE154041 | GEO    | Normal-ized CPM-like | 71 | Short vs. Long OS | limma      | Recurrent GBM survival; discovery cohort |

| Dataset   | Source | Data Type         | n   | Primary Contrast            | DEA Method   | Rationale                                       |
|-----------|--------|-------------------|-----|-----------------------------|--------------|-------------------------------------------------|
| GSE249282 | GEO    | Raw counts        | 58  | Short vs. Long OS           | DESeq2       | DC vaccine-treated GBM; immunotherapy context   |
| GSE186332 | GEO    | Raw counts        | 57  | Responder vs. Non-responder | DESeq2       | Treatment response signatures                   |
| GSE222515 | GEO    | TPM               | 79  | Primary vs. Recurrent       | limma        | Recurrence-associated transcriptional changes   |
| GSE79671  | GEO    | Raw counts        | 35  | Short vs. Long OS           | DESeq2/limma | Independent survival validation                 |
| GSE147352 | GEO    | DESeq2-normalized | 100 | GBM vs. Normal Brain        | limma        | Tumor-intrinsic expression reference            |
| TCGA-GBM  | GDC    | Raw STAR counts   | 280 | Short vs. Long OS           | DESeq2       | Largest GBM RNA-seq cohort; external validation |

**Table 2:** HR > 1 indicates higher expression associated with shorter survival. *padj* = Benjamini–Hochberg adjusted p-value. No genes reach *padj* < 0.05. **Table 3. Top GSEA KEGG Pathways Differentiating Short- from Long-Survival GBM (FDR < 0.05)**

| Gene   | HR   | 95% CI    | <i>p</i> -value | <i>padj</i> | Role                     |
|--------|------|-----------|-----------------|-------------|--------------------------|
| HMGA2  | 1.09 | 1.04–1.15 | 0.0012          | 0.11        | Chromatin architecture   |
| CCR7   | 1.18 | 1.06–1.32 | 0.0022          | 0.11        | Chemokine receptor       |
| HOXC4  | 1.13 | 1.03–1.25 | 0.0091          | 0.23        | HOX transcription factor |
| CAMKK2 | 1.33 | 1.06–1.66 | 0.012           | 0.23        | Calcium signaling        |
| SDC1   | 1.11 | 1.02–1.20 | 0.013           | 0.23        | Proteoglycan             |

| Gene     | HR   | 95% CI    | <i>p</i> -value | <i>p</i> <i>adj</i> | Role                   |
|----------|------|-----------|-----------------|---------------------|------------------------|
| DLG4     | 1.25 | 1.04–1.51 | 0.017           | 0.23                | Synaptic scaffold      |
| SERPINE1 | 1.10 | 1.02–1.19 | 0.018           | 0.23                | PAI-1, ECM remodeling  |
| RPL7A    | 0.81 | 0.67–0.97 | 0.020           | 0.23                | Ribosomal (protective) |
| RPL32    | 0.81 | 0.67–0.97 | 0.020           | 0.23                | Ribosomal (protective) |

**Table 3:** NES = normalized enrichment score. Positive NES indicates enrichment in short-survival GBM; negative NES indicates enrichment in long-survival GBM. **Table 4. Top PPI Network Hub Genes by Degree Centrality**

| Pathway                                           | NES   | <i>p</i> <i>adj</i>  | Direction               |
|---------------------------------------------------|-------|----------------------|-------------------------|
| One carbon pool by folate (hsa00670)              | 2.15  | $5.5 \times 10^{-4}$ | Short-survival enriched |
| Ribosome (hsa03010)                               | −2.33 | $1.8 \times 10^{-8}$ | Long-survival enriched  |
| Coronavirus disease (hsa05171)                    | −2.25 | $1.8 \times 10^{-8}$ | Long-survival enriched  |
| Staphylococcus aureus infection (hsa05150)        | −2.24 | $2.4 \times 10^{-5}$ | Long-survival enriched  |
| Systemic lupus erythematosus (hsa05322)           | −2.12 | $2.5 \times 10^{-5}$ | Long-survival enriched  |
| Hematopoietic cell lineage (hsa04640)             | −2.09 | $1.1 \times 10^{-4}$ | Long-survival enriched  |
| Cytokine–cytokine receptor interaction (hsa04060) | −2.07 | $2.5 \times 10^{-5}$ | Long-survival enriched  |
| Leishmaniasis (hsa05140)                          | −2.05 | $7.3 \times 10^{-5}$ | Long-survival enriched  |
| Complement and coagulation cascades (hsa04610)    | −2.01 | $5.8 \times 10^{-4}$ | Long-survival enriched  |

**Table 4:** Meta  $\log_2$ FC = pooled  $\log_2$  fold-change from random-effects meta-analysis (negative = downregulated in short-survival GBM). **Table 5. LASSO-Cox Regression Coefficients (31-Gene Signature)**

| Gene  | Degree | Betweenness | Meta $\log_2$ FC | Function                     |
|-------|--------|-------------|------------------|------------------------------|
| SRC   | 91     | 0.070       | −0.30            | Non-receptor tyrosine kinase |
| H4C11 | 62     | 0.043       | −1.92            | Histone H4 variant           |
| CSF1R | 56     | 0.016       | −0.48            | Macrophage CSF receptor      |

| Gene   | Degree | Betweenness | Meta log <sub>2</sub> FC | Function                |
|--------|--------|-------------|--------------------------|-------------------------|
| APOE   | 54     | 0.027       | −0.43                    | Apolipoprotein E        |
| CYBB   | 54     | 0.019       | −0.45                    | NADPH oxidase subunit   |
| PPP1CA | 54     | 0.035       | −0.25                    | Protein phosphatase 1   |
| CD86   | 52     | 0.011       | −2.18                    | Co-stimulatory molecule |
| BTK    | 51     | 0.021       | −0.50                    | Bruton tyrosine kinase  |
| PIK3CG | 50     | 0.025       | −0.49                    | PI3K gamma              |
| TYROBP | 48     | 0.008       | −0.57                    | DAP12 adaptor protein   |

**Table 5:** Positive coefficient = risk-associated (higher expression associated with shorter survival). Negative coefficient = protective. **Table 6. Immune Cell Infiltration Correlations with Risk Score (TCGA-GBM, n = 285)**

| Gene      | Coefficient | Direction  |
|-----------|-------------|------------|
| HOXC4     | 0.251       | Risk       |
| VAT1L     | 0.183       | Risk       |
| CAMKK2    | 0.170       | Risk       |
| HMGA2     | 0.163       | Risk       |
| SDC1      | 0.114       | Risk       |
| TSFM      | 0.107       | Risk       |
| RIC8A     | 0.083       | Risk       |
| PNPLA7    | 0.082       | Risk       |
| BHMT2     | 0.077       | Risk       |
| PCDHA11   | 0.065       | Risk       |
| TNFRSF12A | 0.051       | Risk       |
| OTOA      | 0.038       | Risk       |
| HHATL     | 0.026       | Risk       |
| LYVE1     | 0.025       | Risk       |
| GSG1      | 0.023       | Risk       |
| OLFML1    | 0.012       | Risk       |
| TPBGL     | 0.009       | Risk       |
| STAC2     | 0.006       | Risk       |
| ZNF831    | 0.002       | Risk       |
| QARS1     | −0.198      | Protective |
| ZKSCAN8   | −0.161      | Protective |
| CD207     | −0.134      | Protective |
| POU4F1    | −0.094      | Protective |
| NPY1R     | −0.088      | Protective |
| CD86      | −0.051      | Protective |
| CAV1      | −0.049      | Protective |
| CSKMT     | −0.047      | Protective |
| CCN5      | −0.043      | Protective |
| NICOL1    | −0.039      | Protective |

| Gene  | Coefficient | Direction  |
|-------|-------------|------------|
| SKIC8 | −0.006      | Protective |
| TKTL1 | −0.004      | Protective |

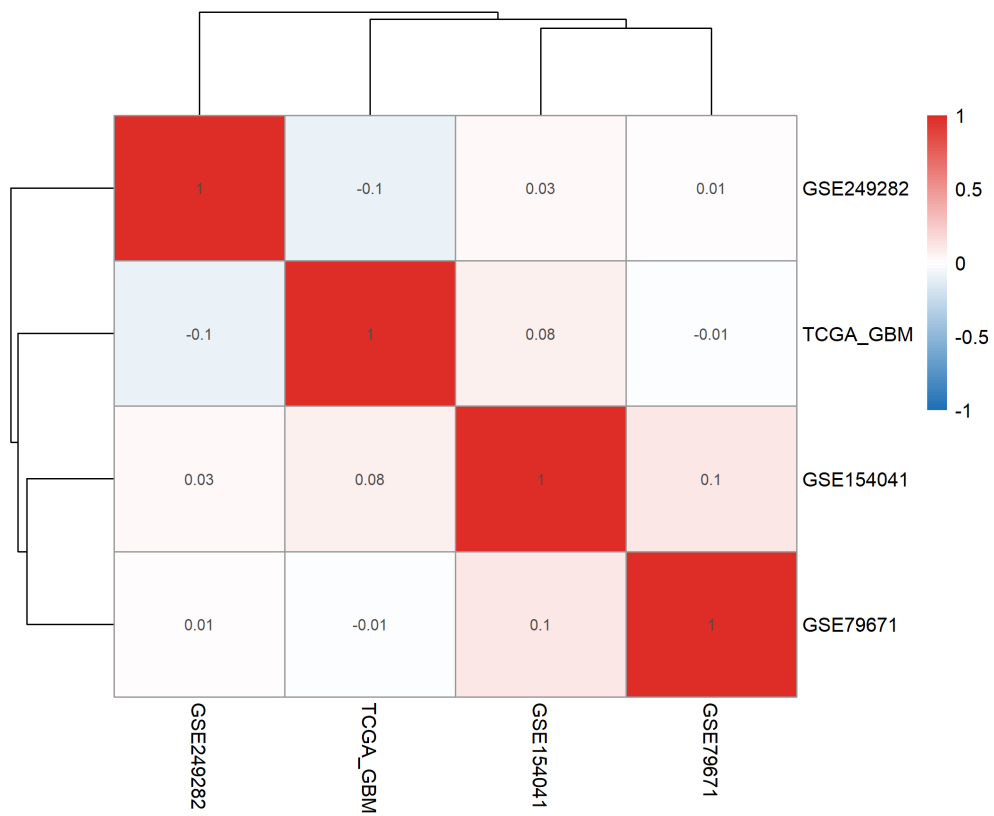
**Table 6:**  $p_{adj}$  = Benjamini–Hochberg adjusted p-value. Significance threshold:  $p_{adj} < 0.05$ . **Table 7. Drug–Gene Interactions for Top Hub Targets**

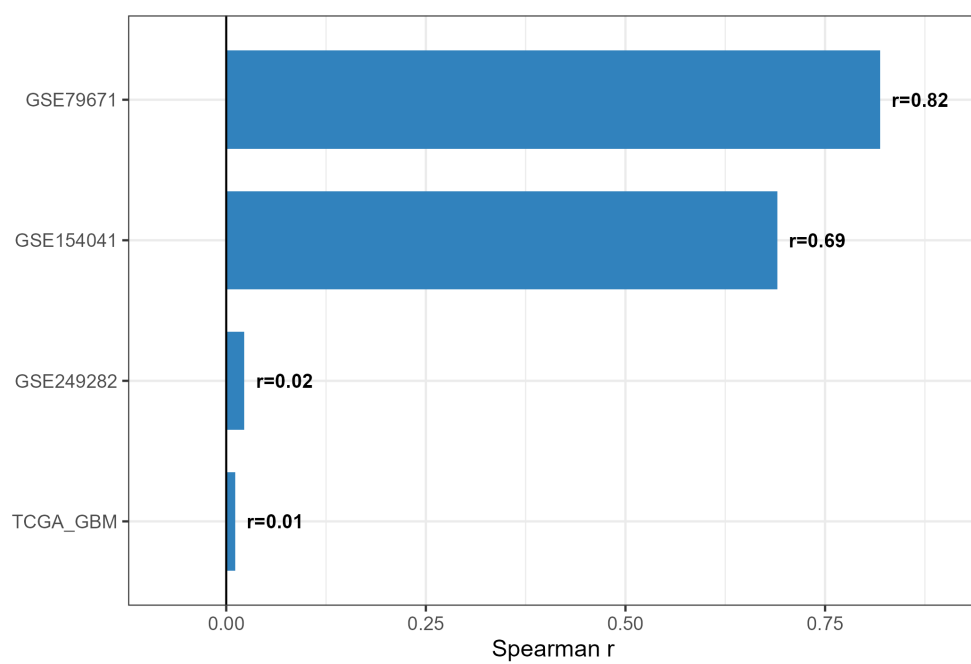
| Cell Type               | Spearman $r$ | $p_{adj}$            | Significant |
|-------------------------|--------------|----------------------|-------------|
| Fibroblast              | 0.25         | $2.5 \times 10^{-4}$ | Yes         |
| Macrophage              | 0.17         | 0.019                | Yes         |
| Neutrophil              | 0.15         | 0.044                | Yes         |
| CD4 <sup>+</sup> T cell | 0.12         | 0.12                 | No          |
| Monocyte                | 0.11         | 0.12                 | No          |
| CD8 <sup>+</sup> T cell | 0.09         | 0.21                 | No          |
| B cell                  | −0.08        | 0.24                 | No          |
| NK cell                 | 0.08         | 0.26                 | No          |
| Mast cell               | −0.06        | 0.31                 | No          |
| Dendritic cell          | 0.06         | 0.31                 | No          |

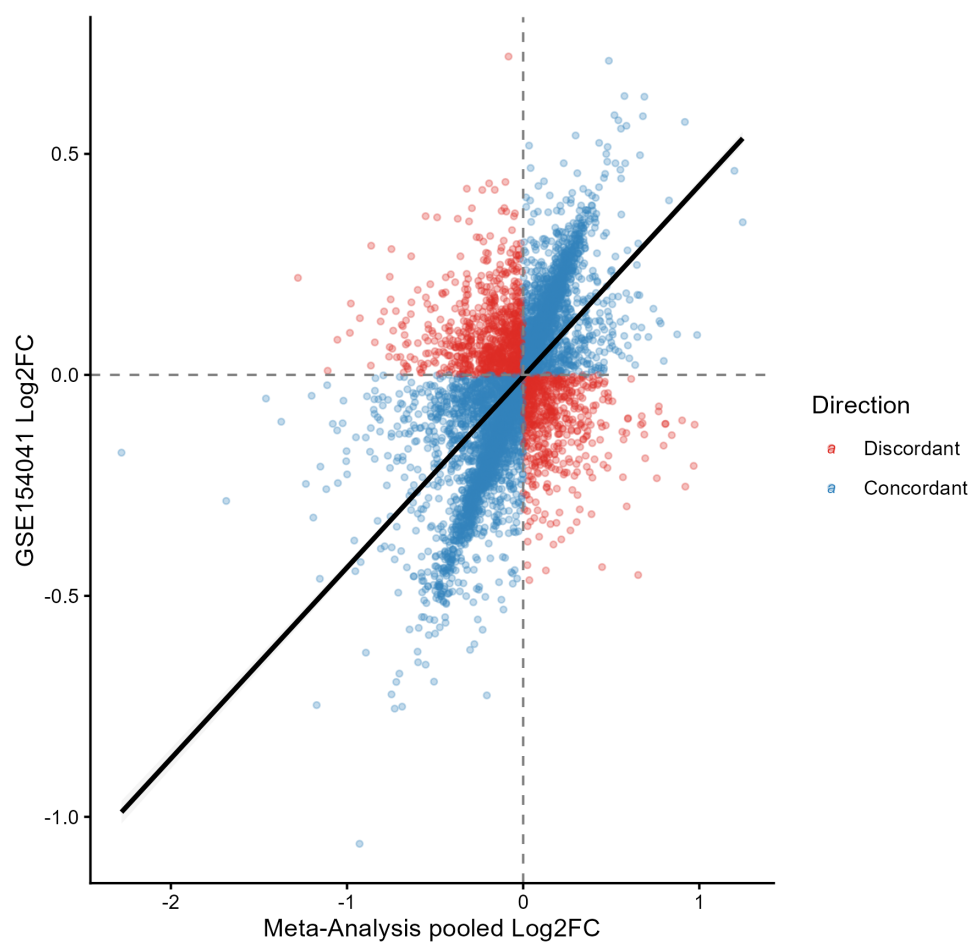
**Table 7:** Drug interactions from DGIdb v4.0. Priority determined by integration of hub status, differential expression, and druggability. # Figures

| Gene     | Total Drugs | Inhibitors | Gene Family                            | Priority |
|----------|-------------|------------|----------------------------------------|----------|
| PIK3CG   | 130         | 45         | PI3 Kinase; Clinically Actionable      | Medium   |
| GRIN2A   | 68          | 32         | Ion Channel; Clinically Actionable     | Medium   |
| APOE     | 58          | 6          | Druggable Genome                       | Medium   |
| CSF1R    | 54          | 22         | Tyrosine Kinase; Clinically Actionable | Medium   |
| SRC      | 54          | 16         | Tyrosine Kinase; Clinically Actionable | Medium   |
| BTK      | 52          | 26         | Tyrosine Kinase; Clinically Actionable | Medium   |
| SERPINE1 | 34          | 1          | Protease Inhibitor                     | Medium   |

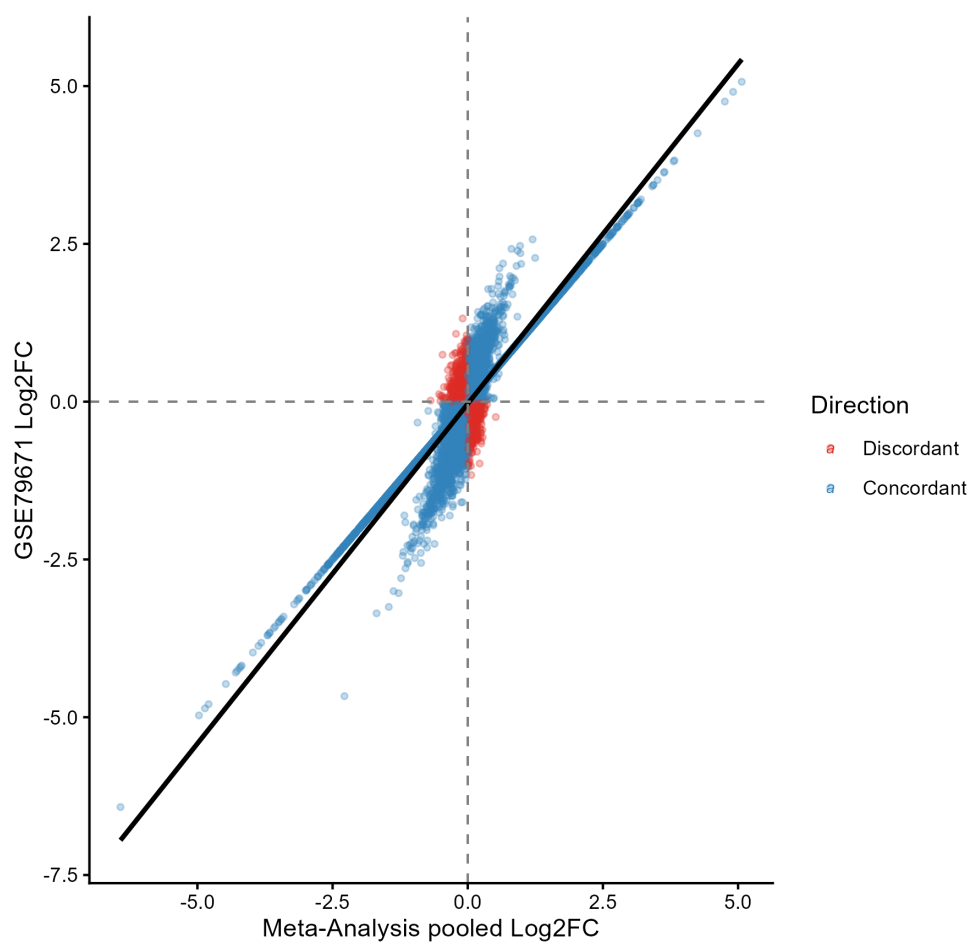
**Figure 1.** Cross-cohort concordance analysis.

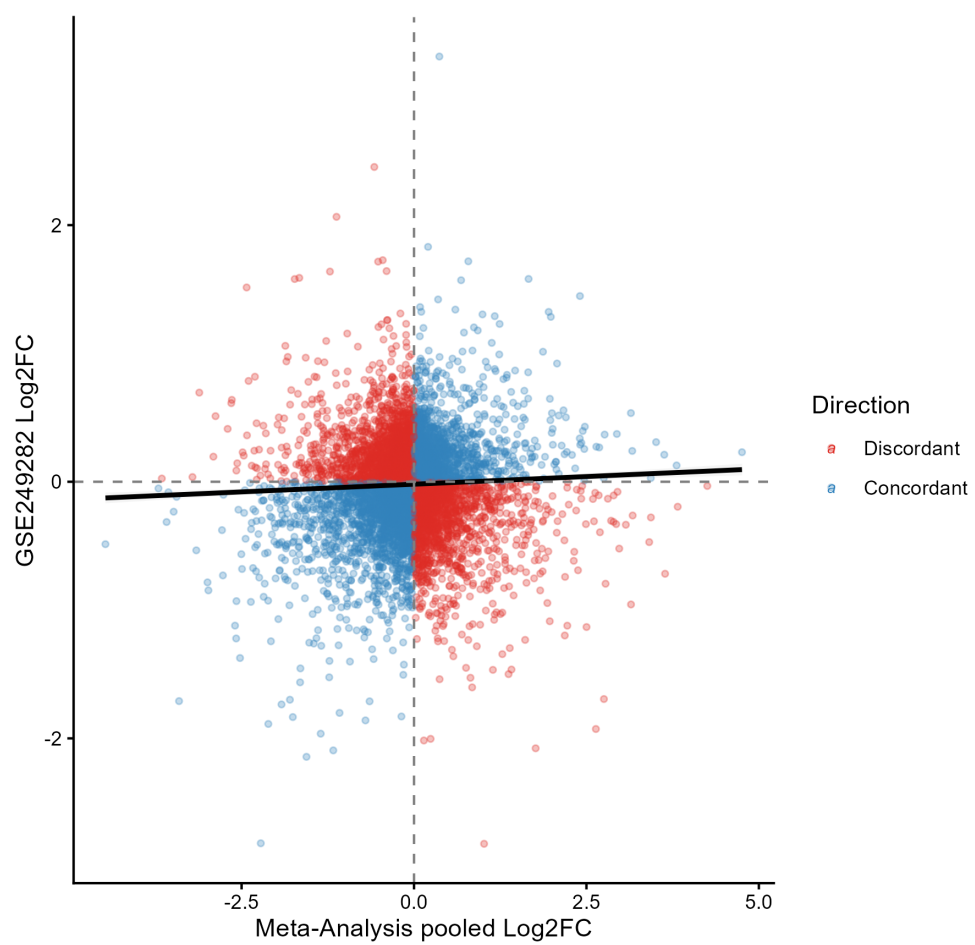


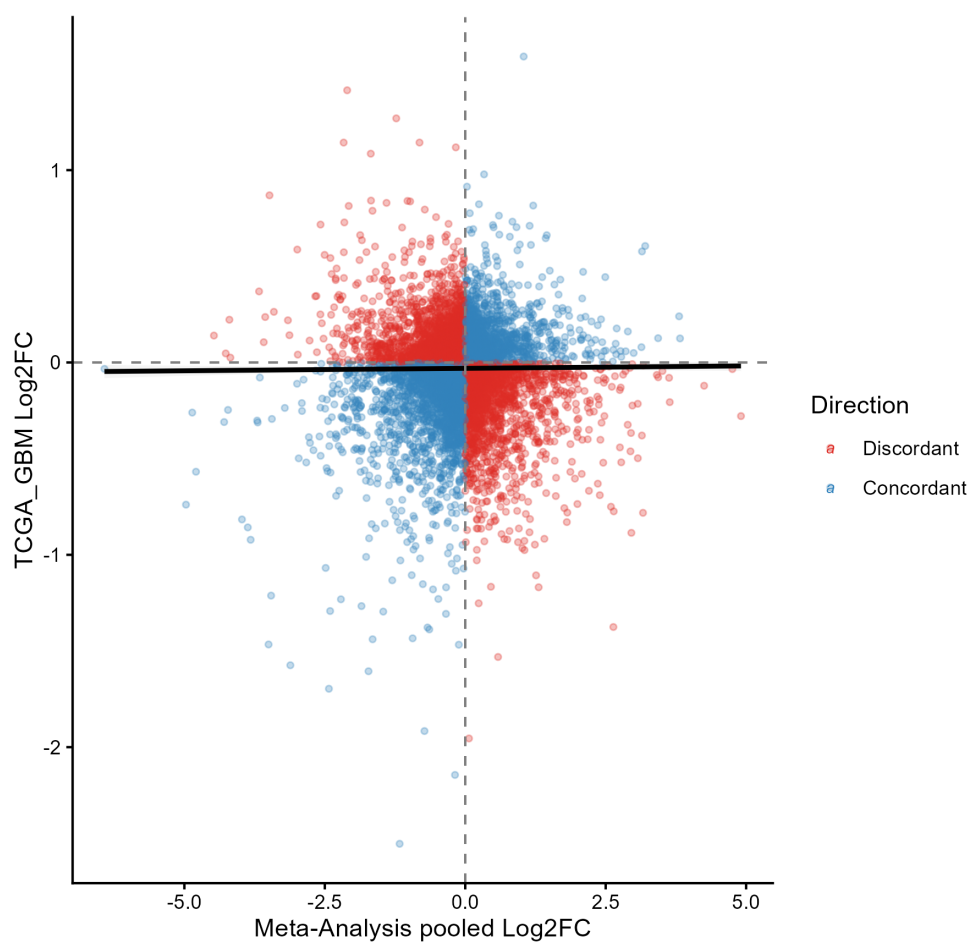




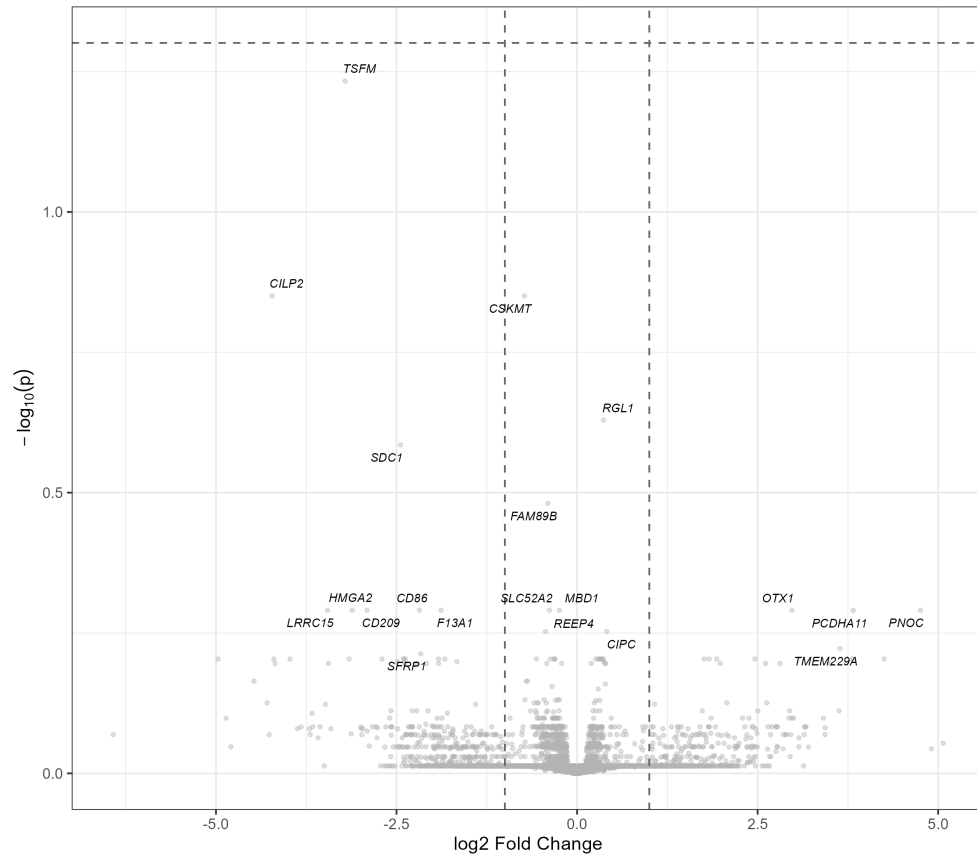






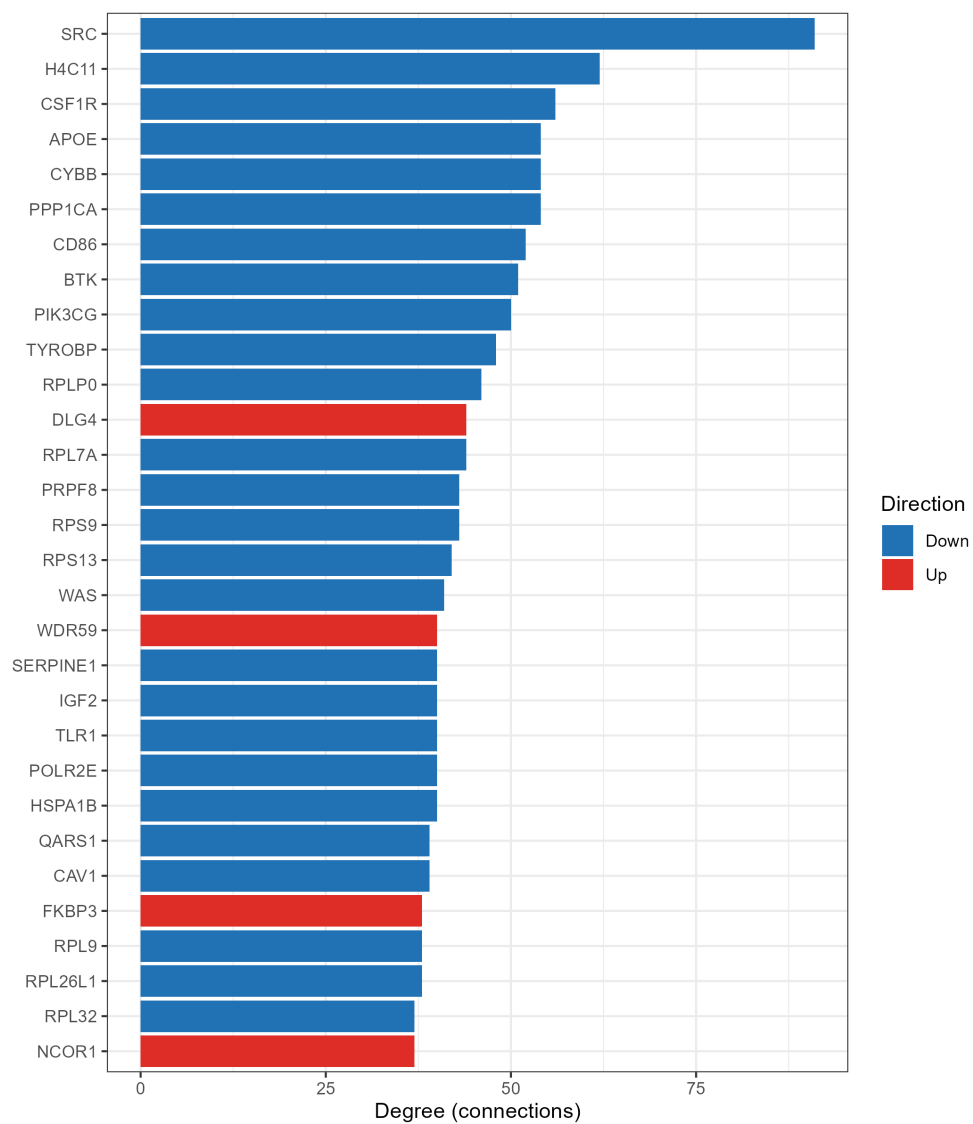


**Figure 2.** Meta-analysis volcano plot.

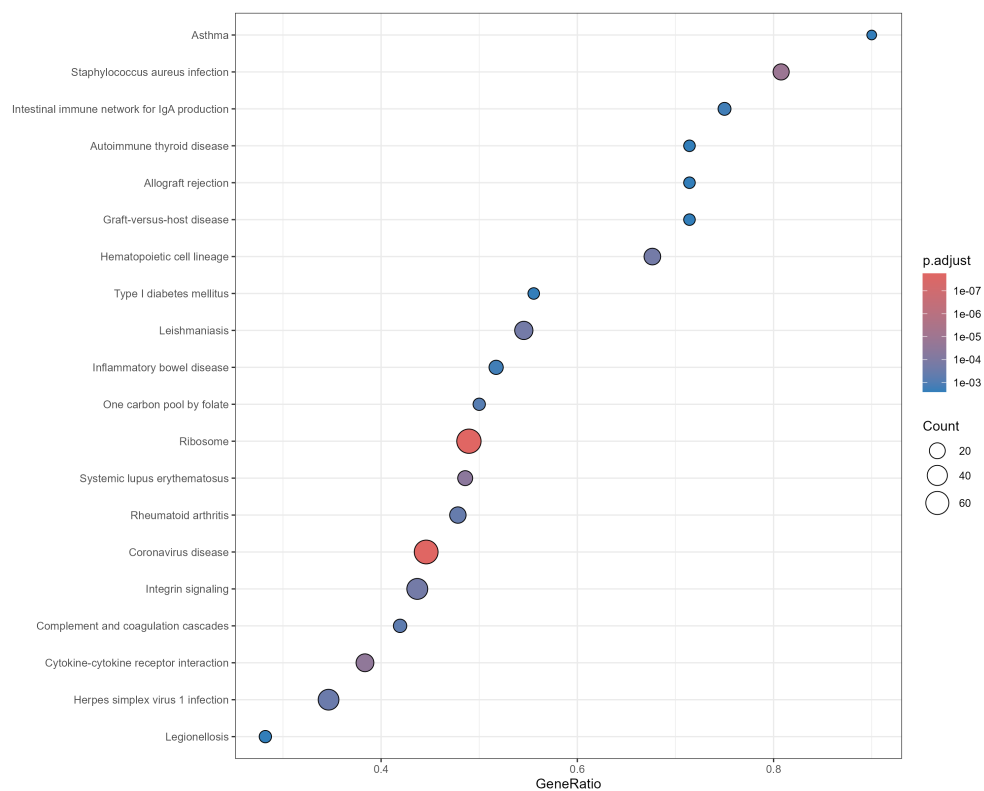


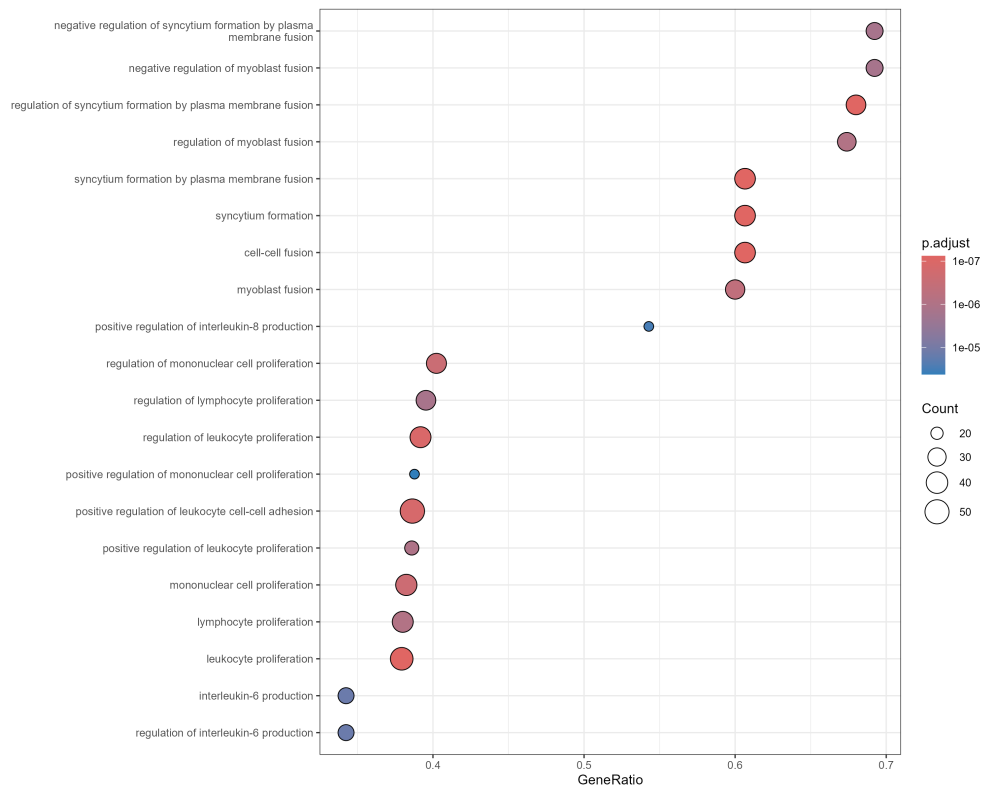
**Figure 3.** PPI network of top 500 meta-analytic genes.



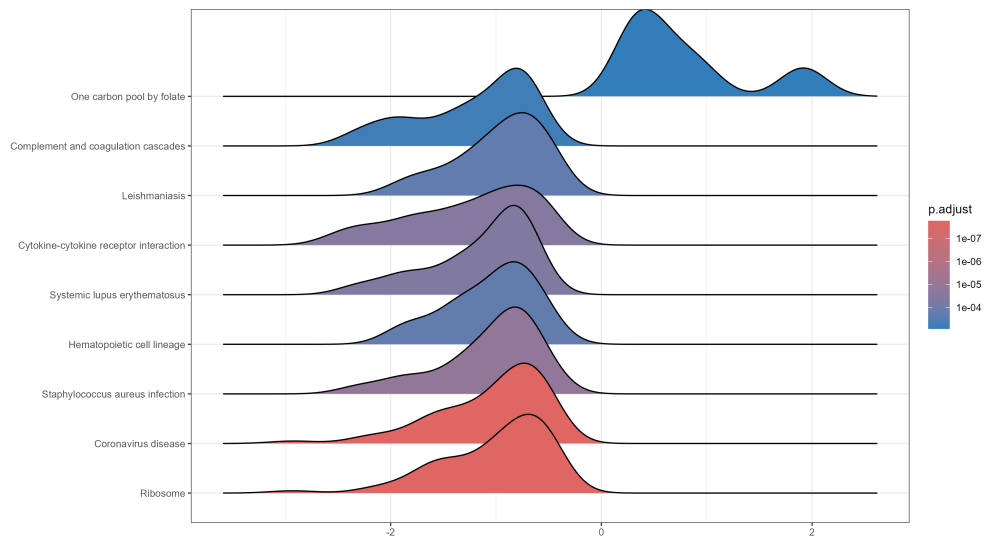


**Figure 5.** GSEA enrichment results.



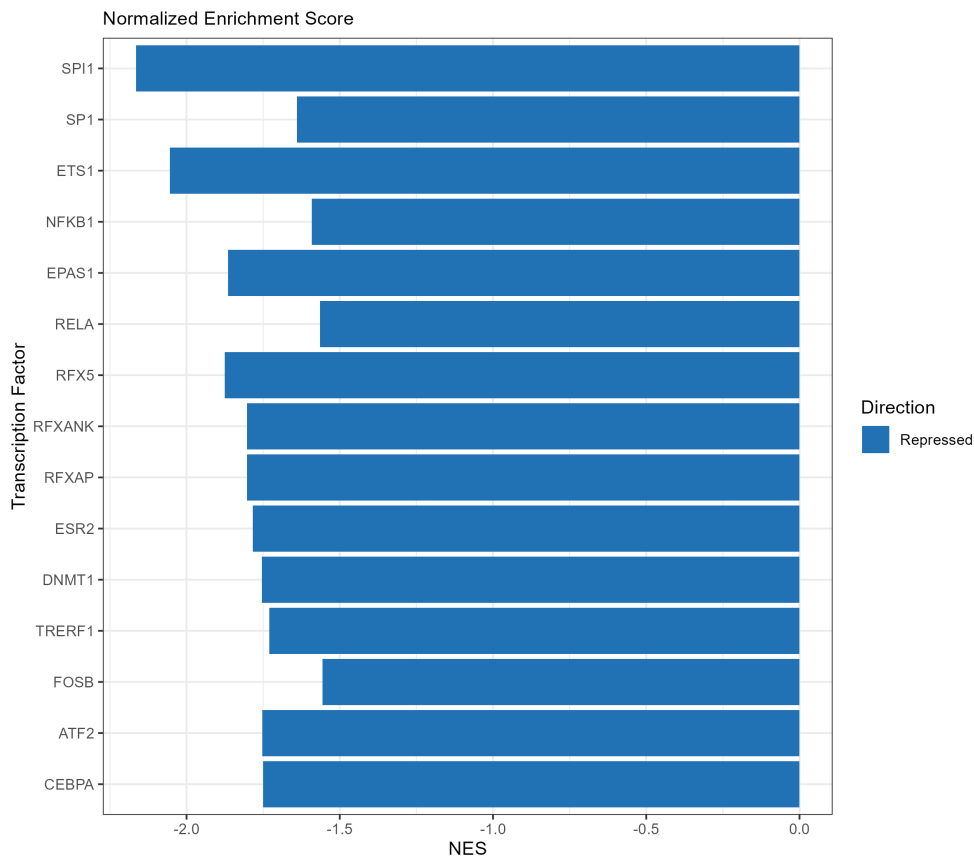


**Figure 6.** GSEA running enrichment score plots.

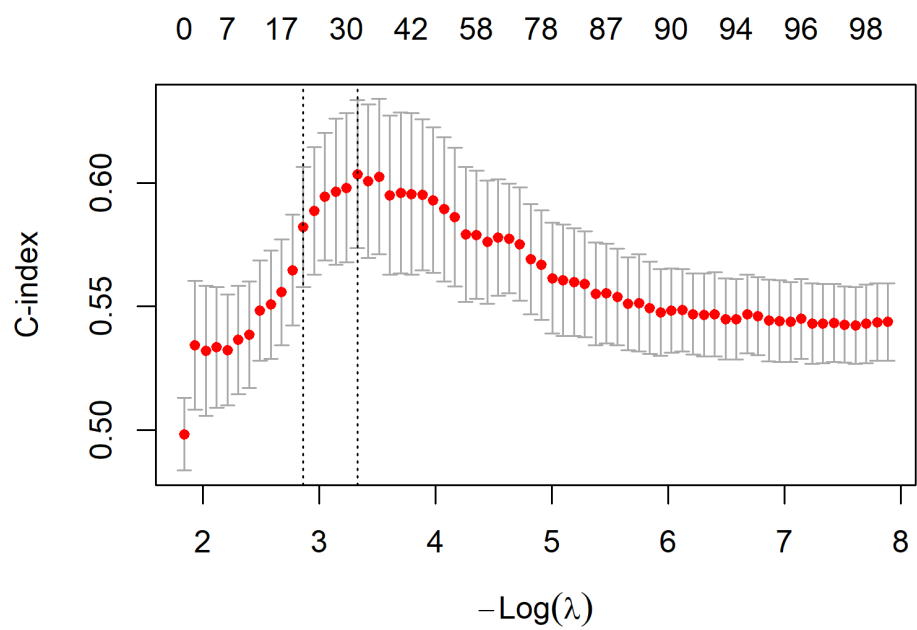


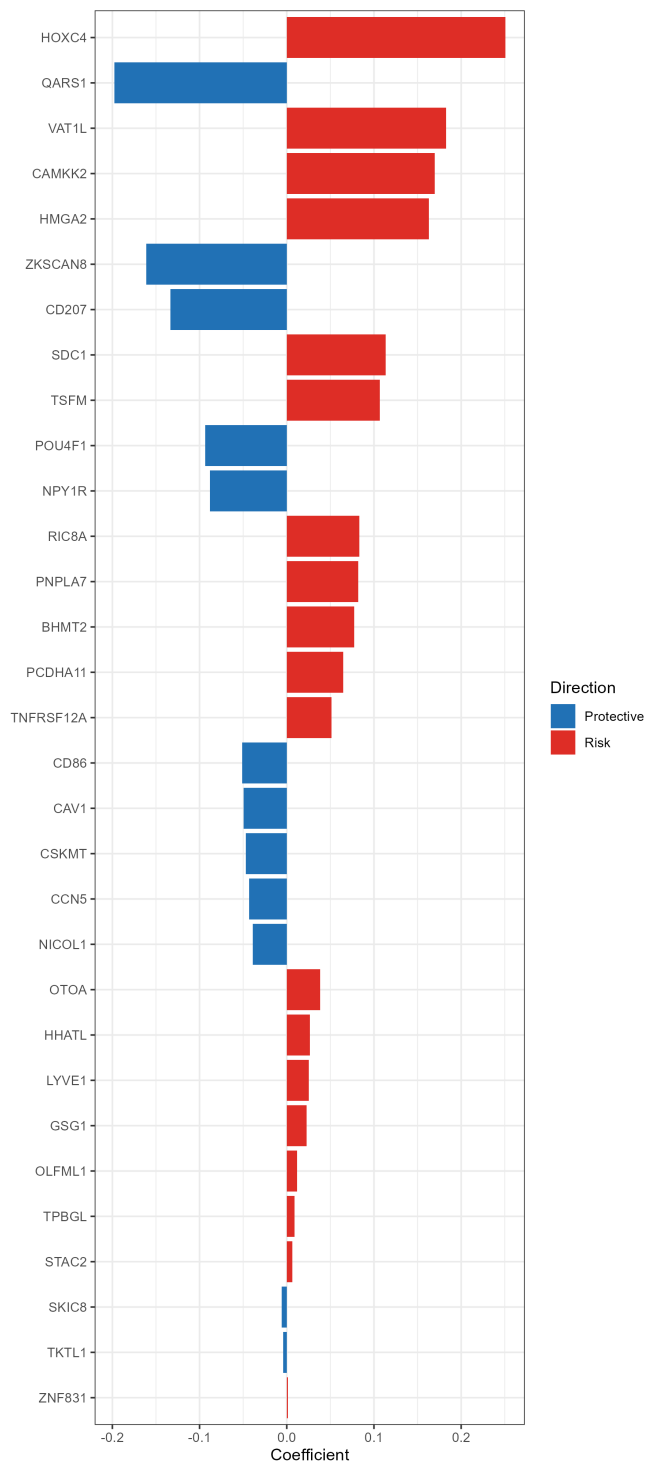


**Figure 7.** Transcription factor enrichment analysis.

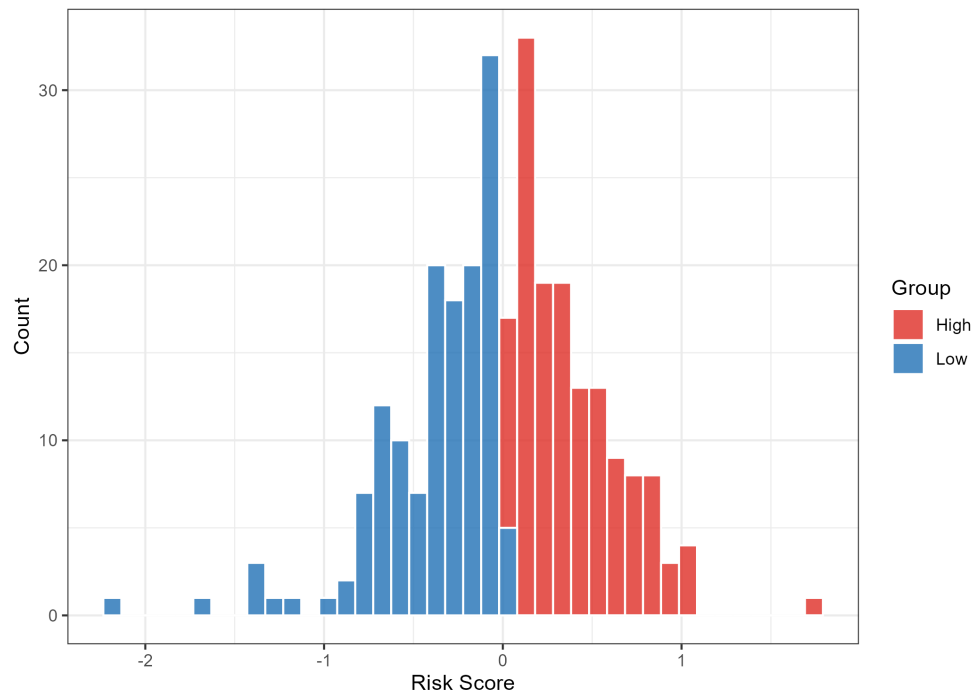


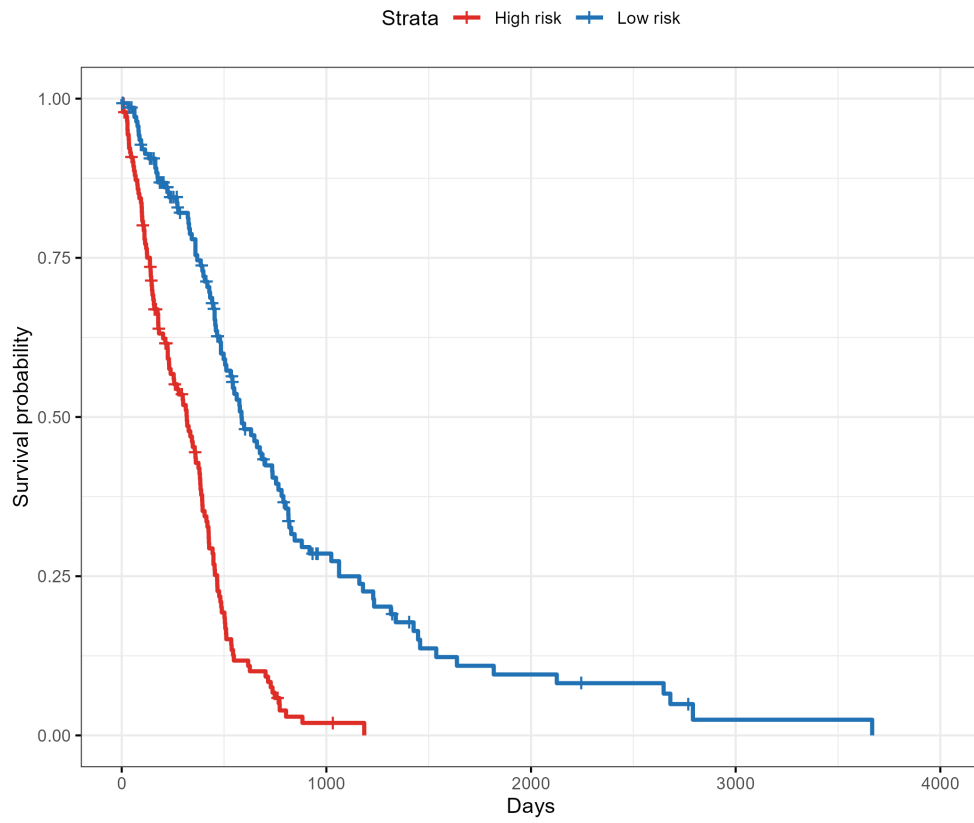
**Figure 8.** LASSO-Cox prognostic signature.



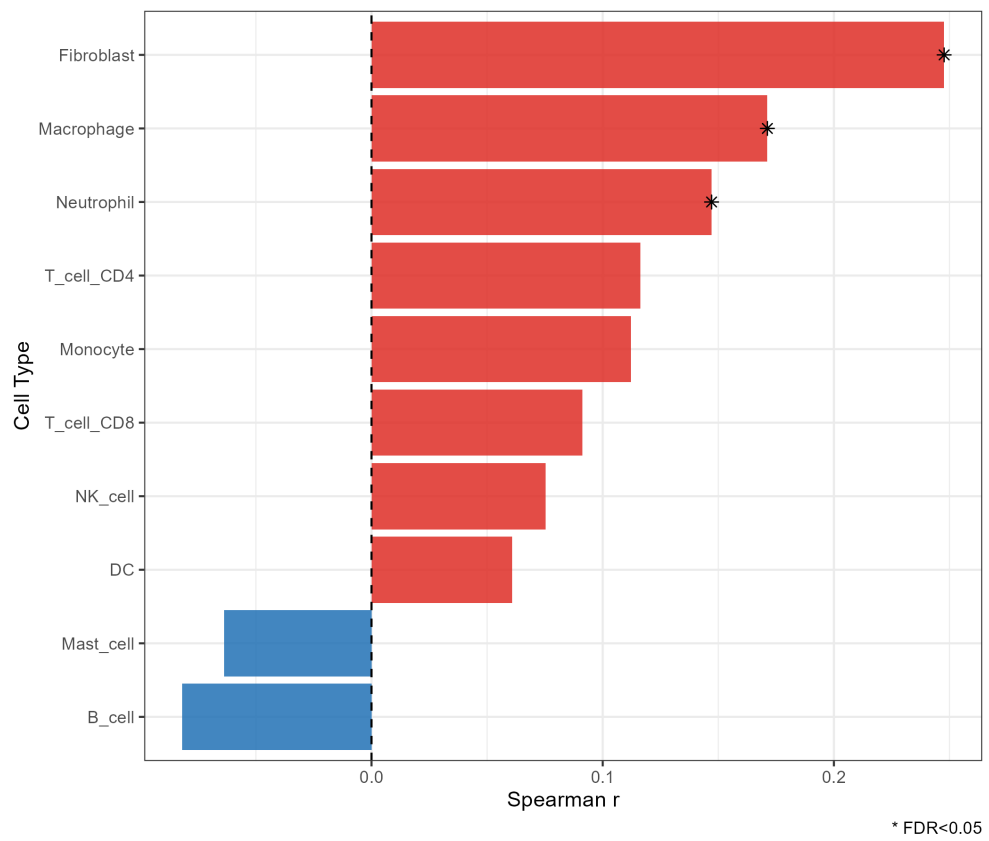
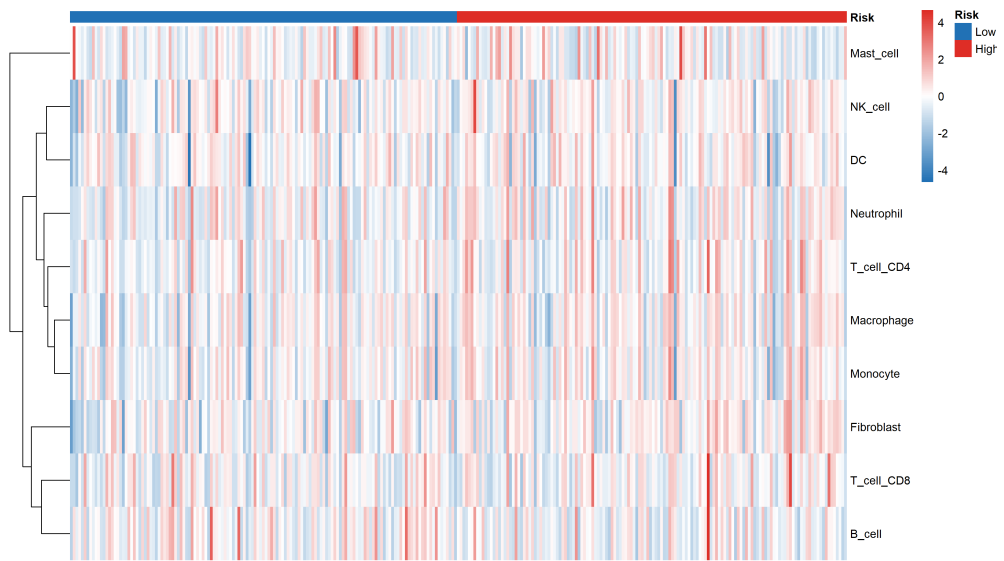


**Figure 9.** Risk score validation.

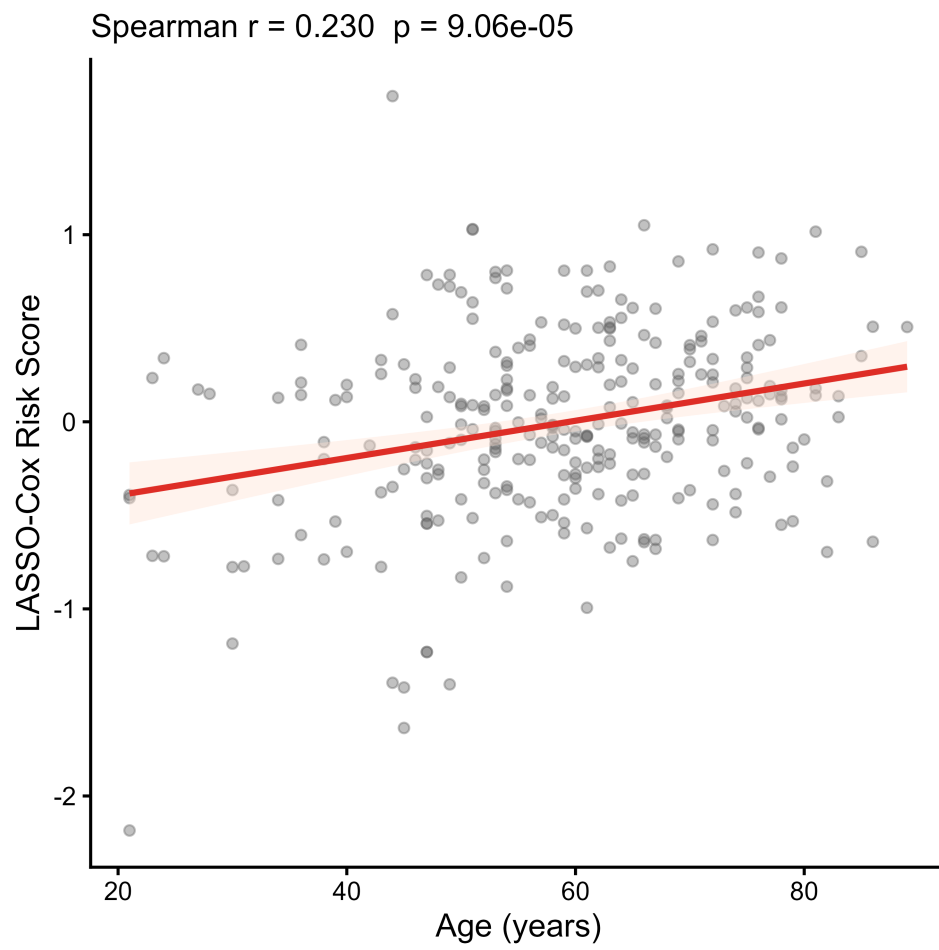


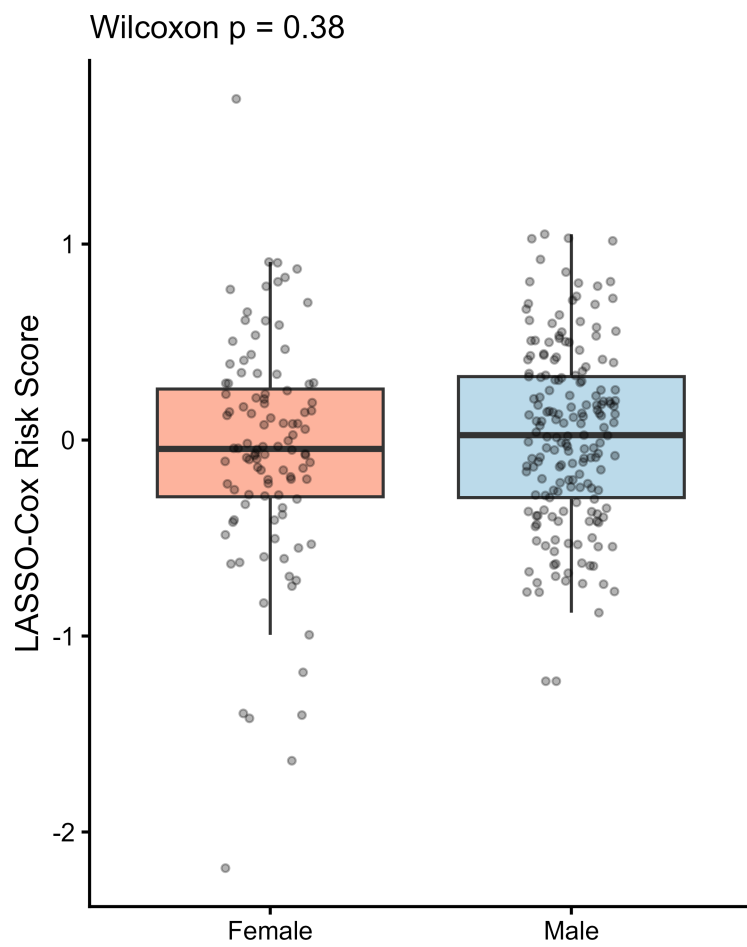


**Figure 10.** Immune microenvironment landscape.

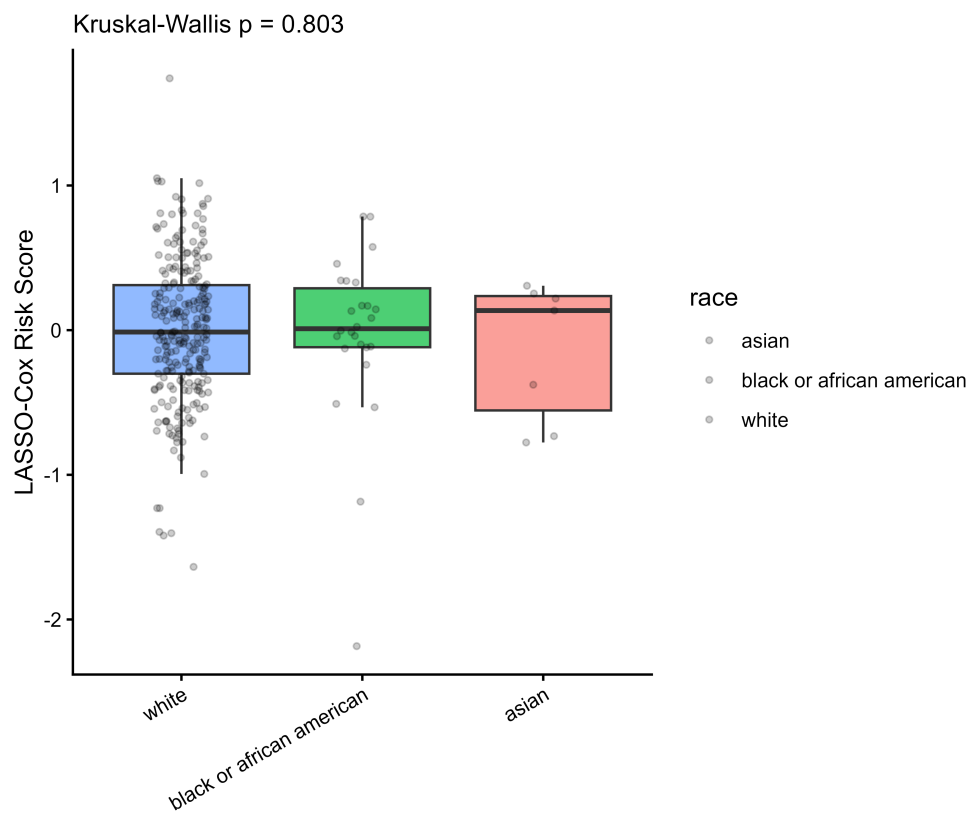


**Figure 11.** Clinical association plots.









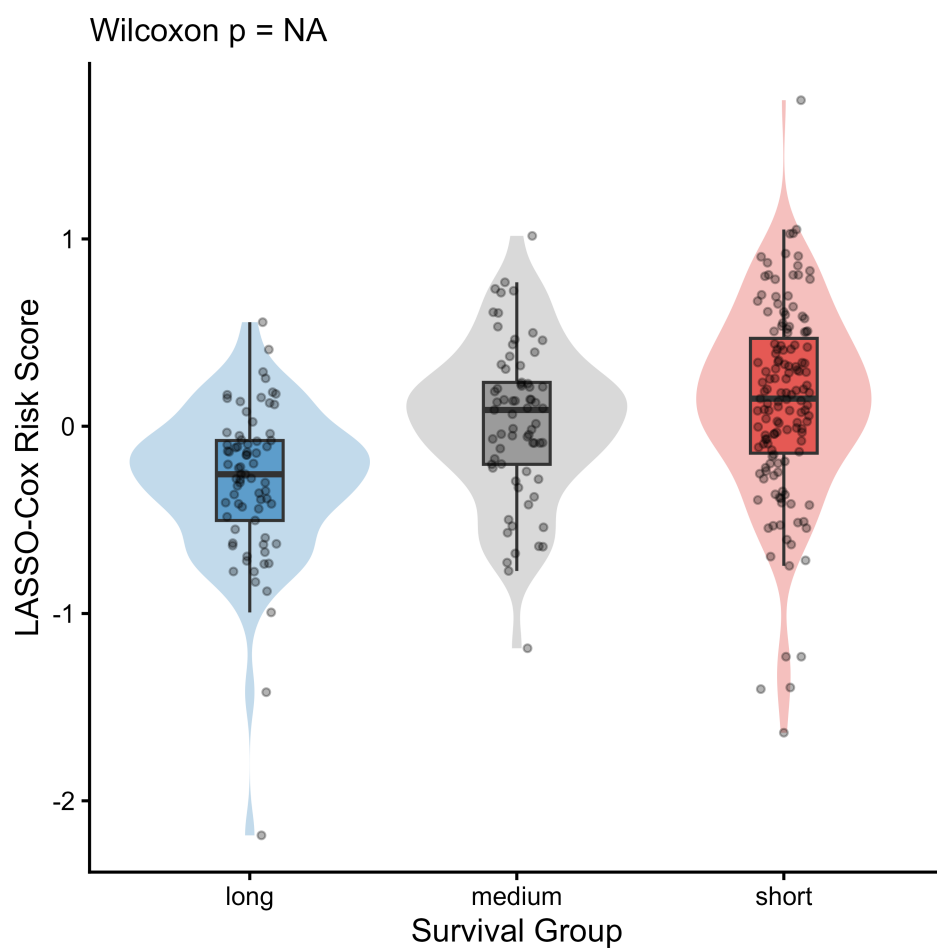


Figure 12. Drug target prioritization.

